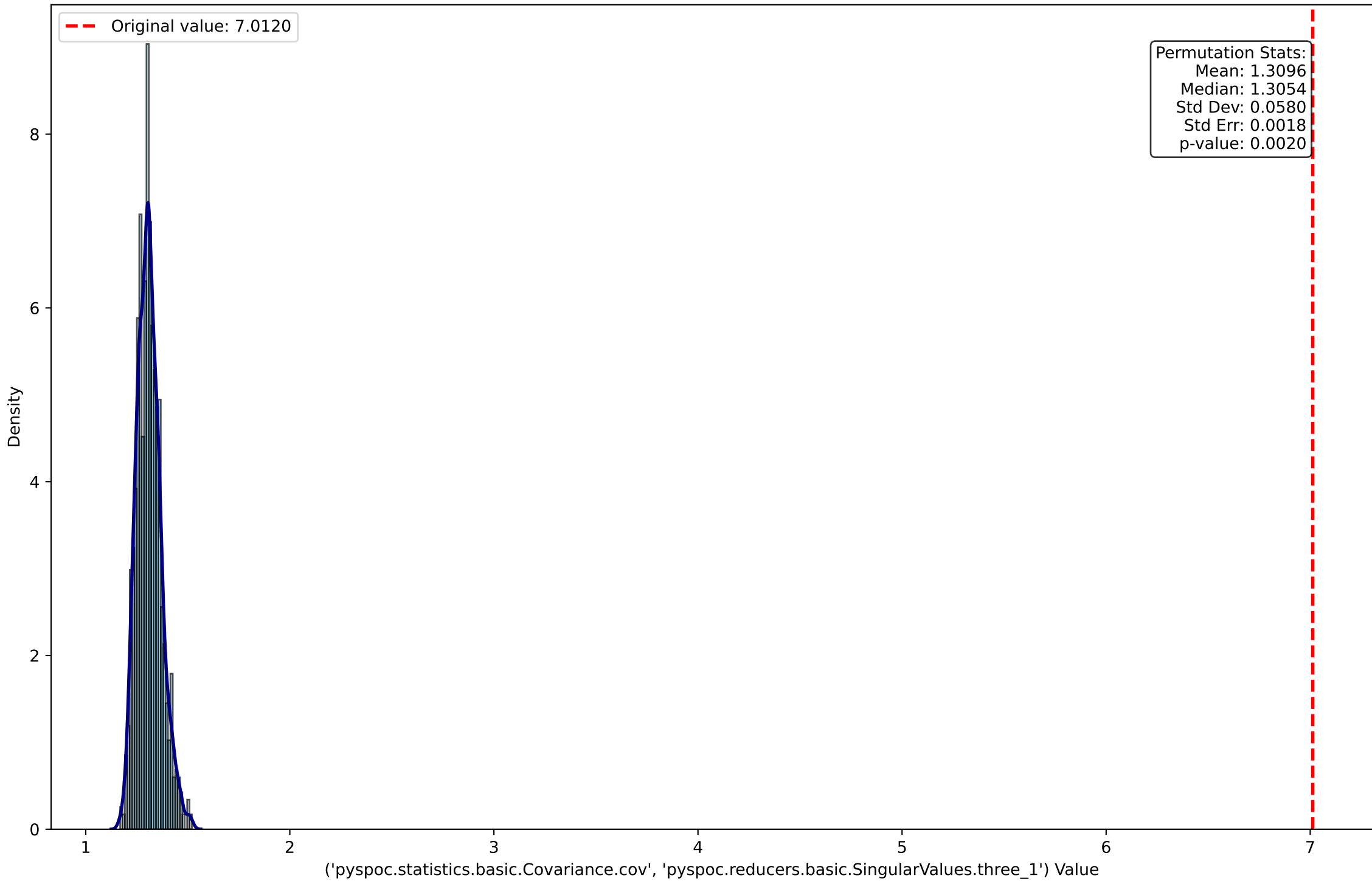
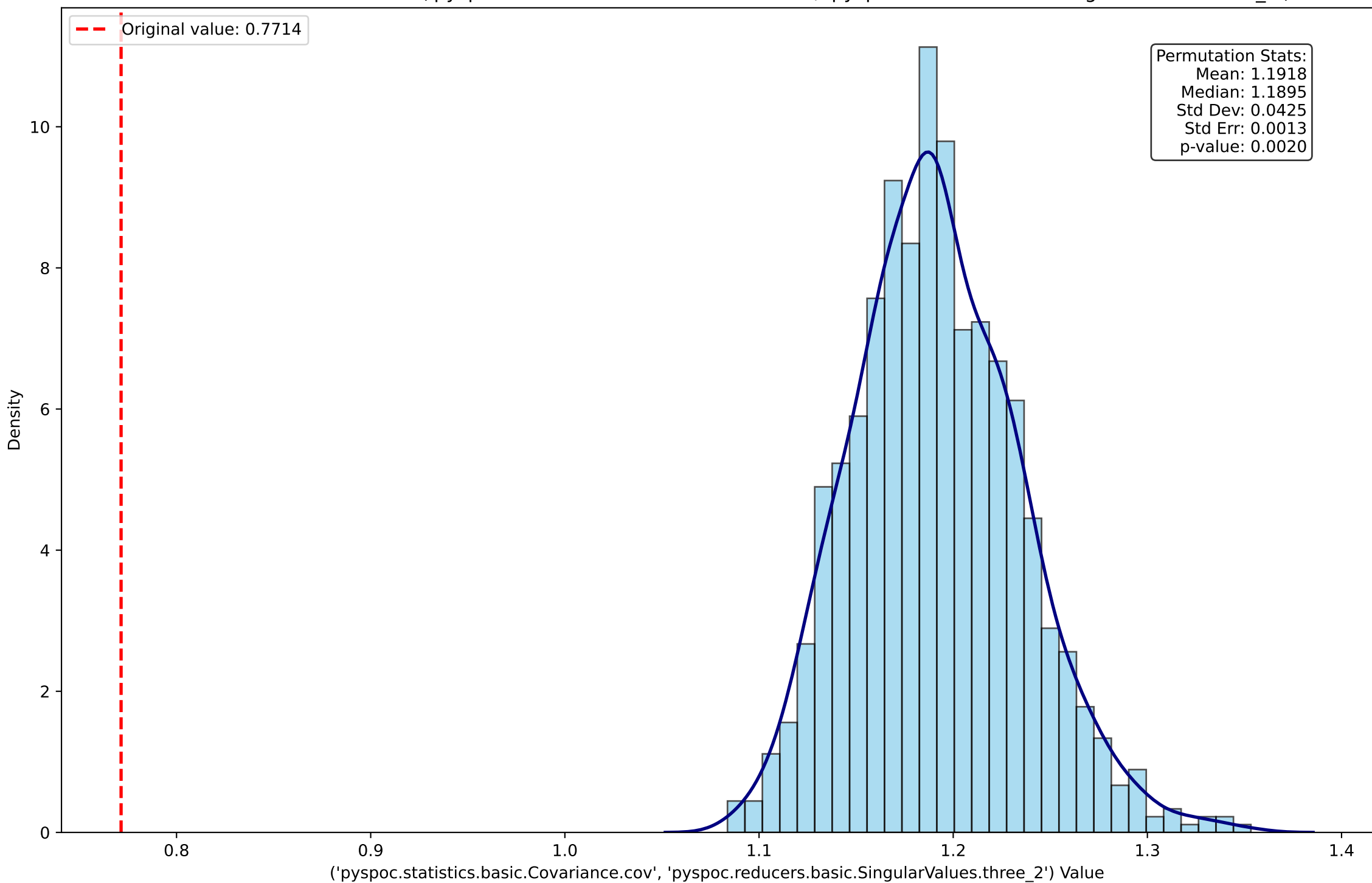


Number of permutations: 1000
Number of bootstrap samples: 1000
Row bootstrap sample fraction: 90.0%
Column bootstrap sample fraction: 90.0%
Number of perturbations: 1000
Scale factor for perturbations: 0.1
Normalization: z-score
Random seed: 42

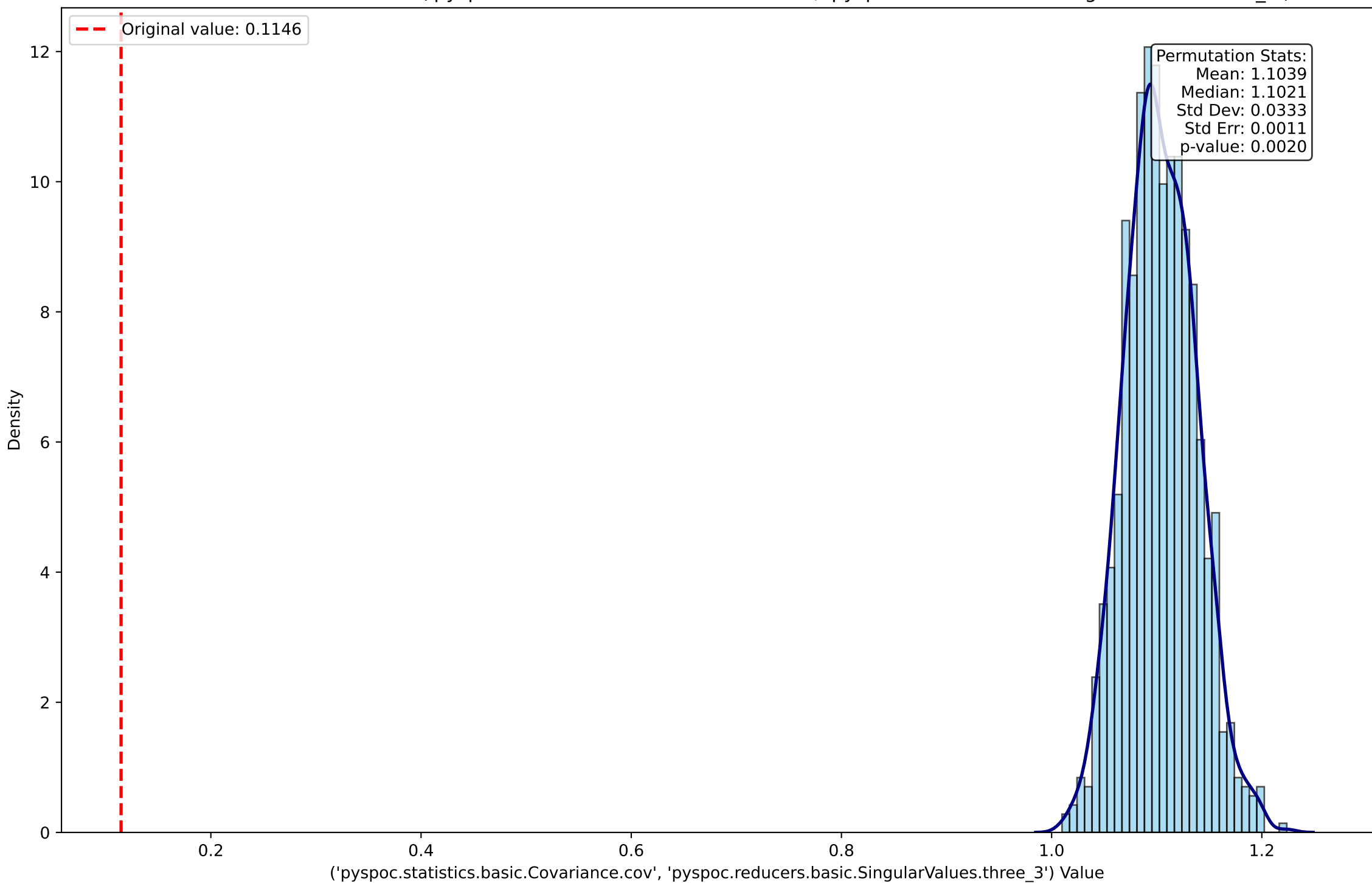
Permutation Distribution for ('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.SingularValues.three_1')



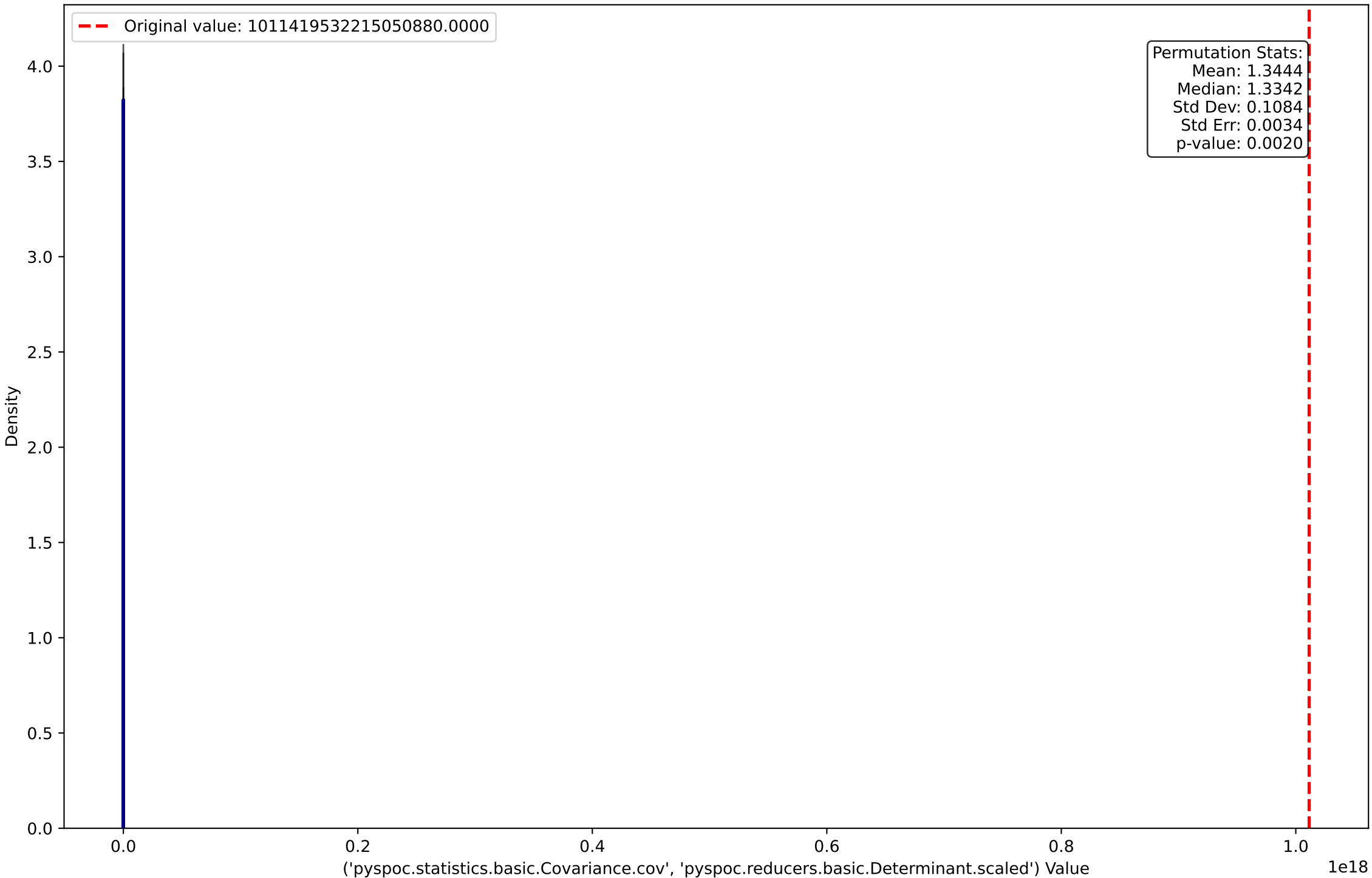
Permutation Distribution for ('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.SingularValues.three_2')



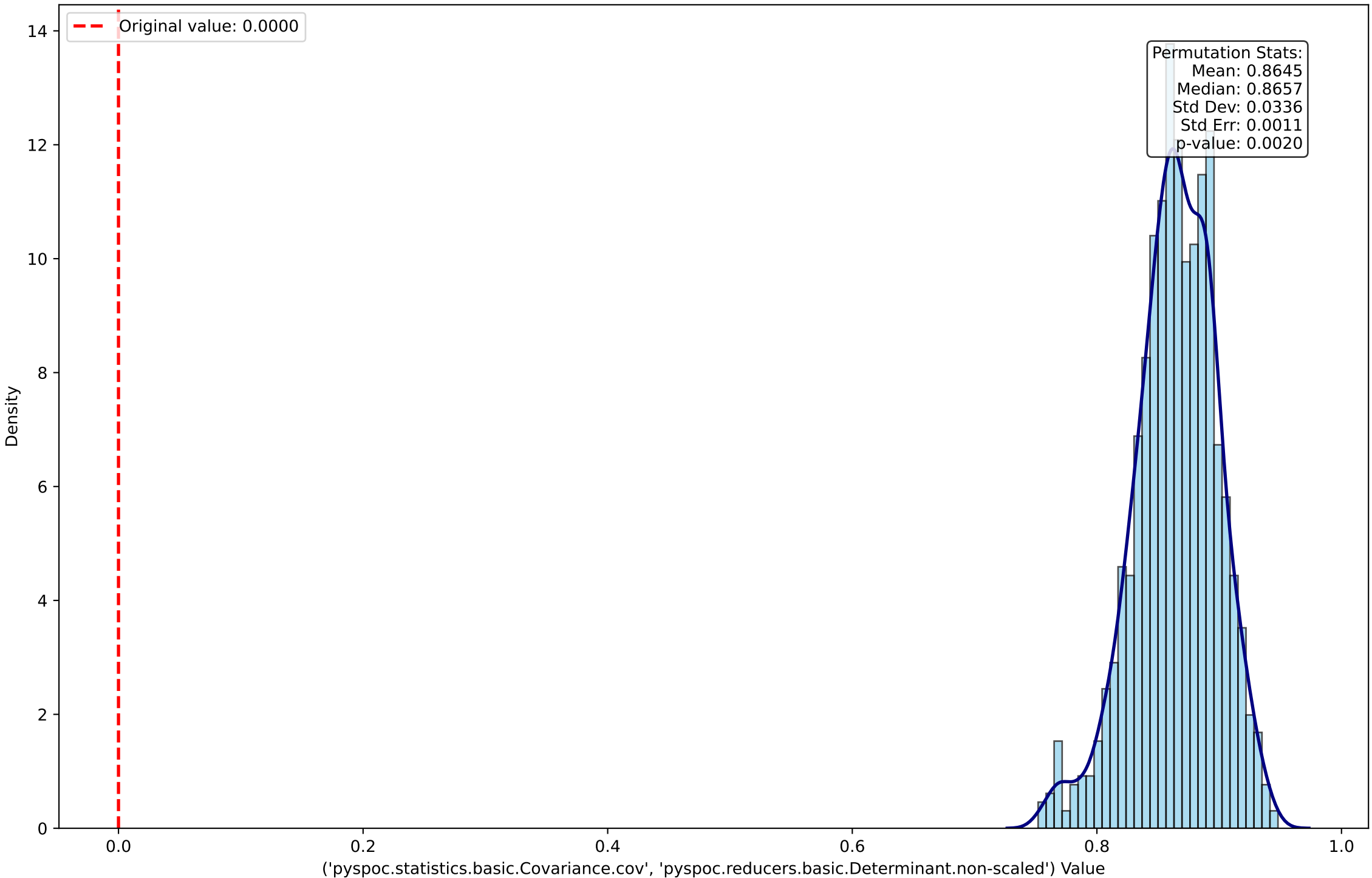
Permutation Distribution for ('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.SingularValues.three_3')



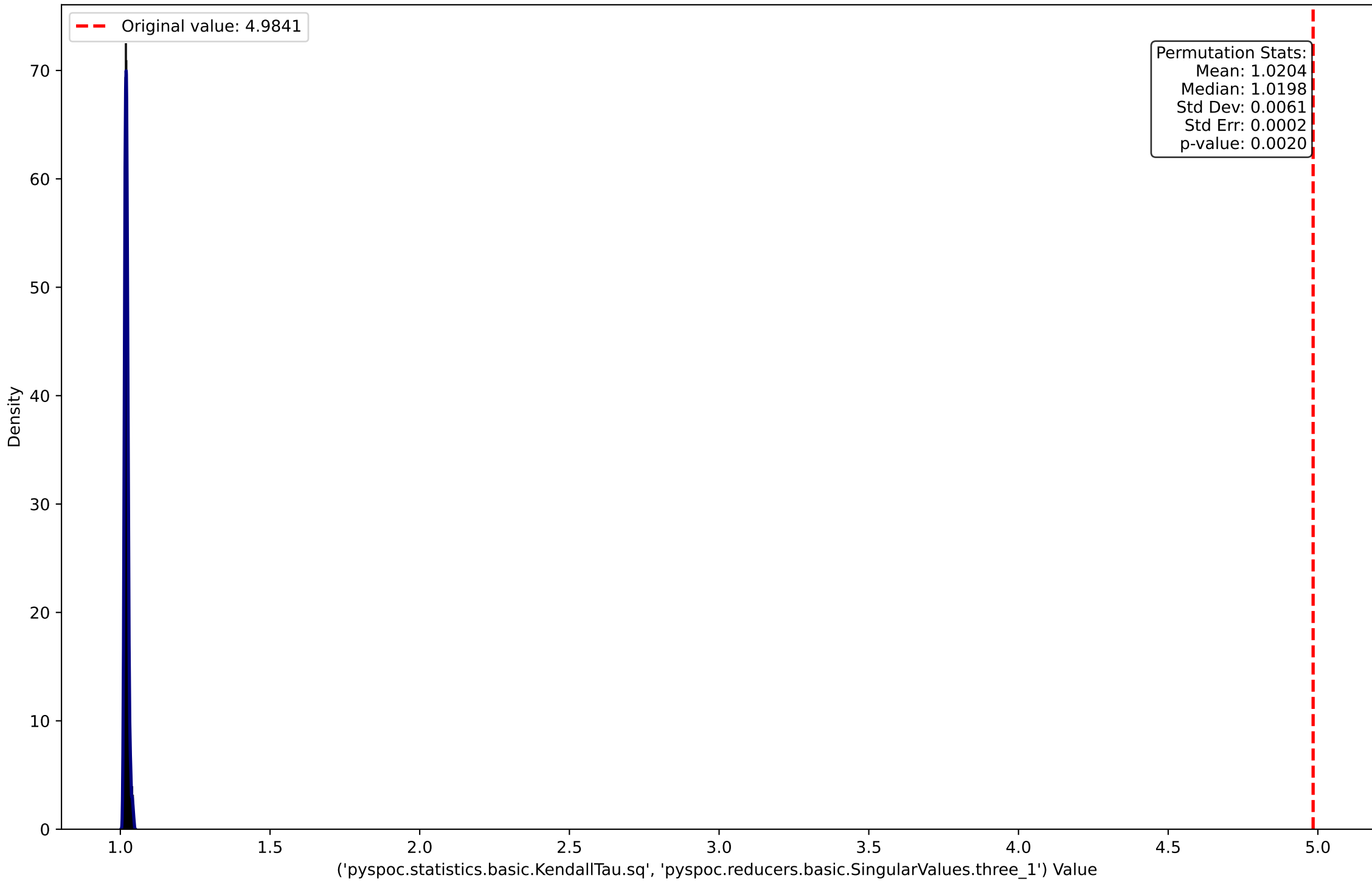
Permutation Distribution for ('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.Determinant.scaled')



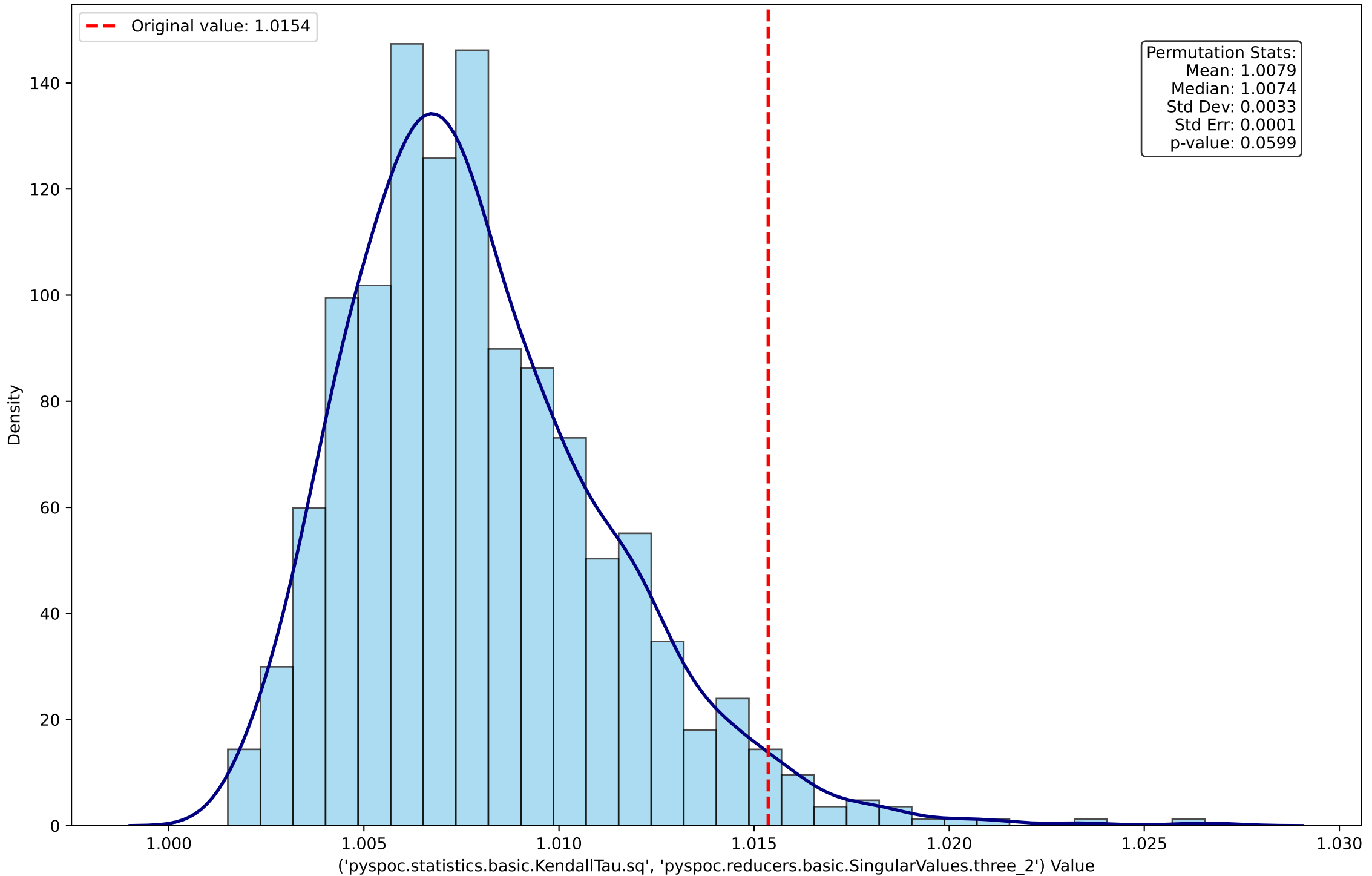
Permutation Distribution for ('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.Determinant.non-scaled')



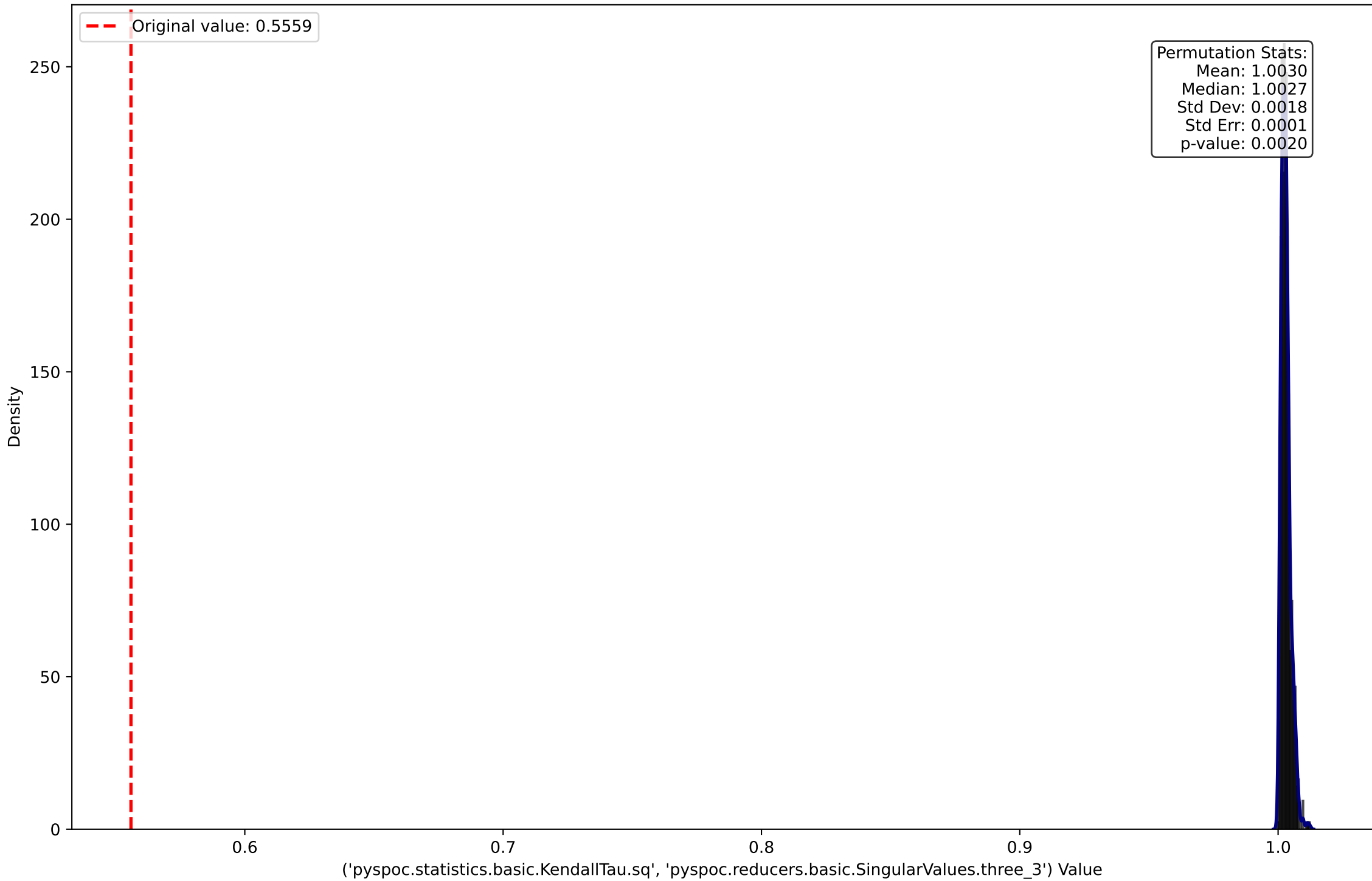
Permutation Distribution for ('pyspoc.statistics.basic.KendallTau.sq', 'pyspoc.reducers.basic.SingularValues.three_1')



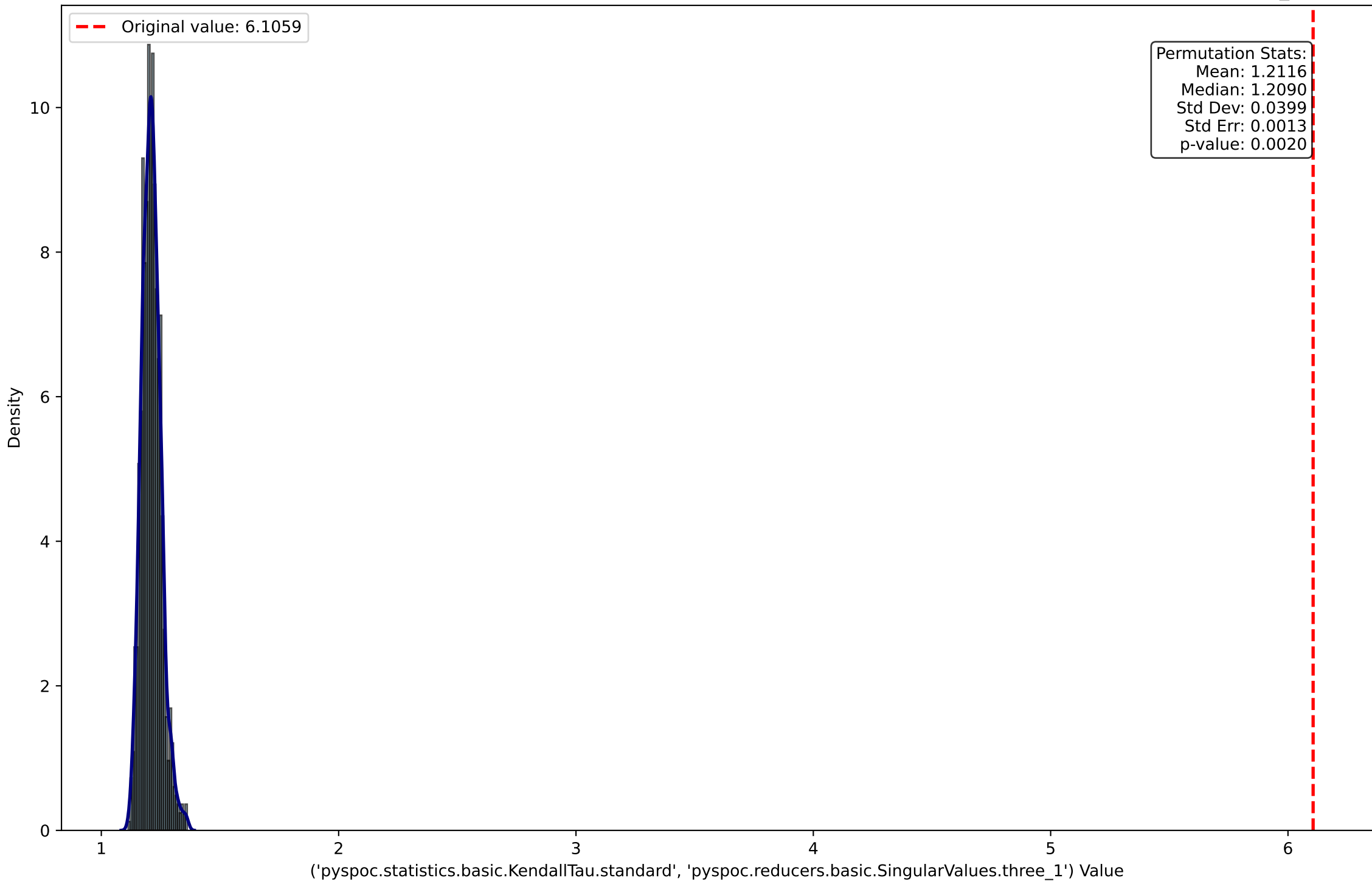
Permutation Distribution for ('pyspoc.statistics.basic.KendallTau.sq', 'pyspoc.reducers.basic.SingularValues.three_2')



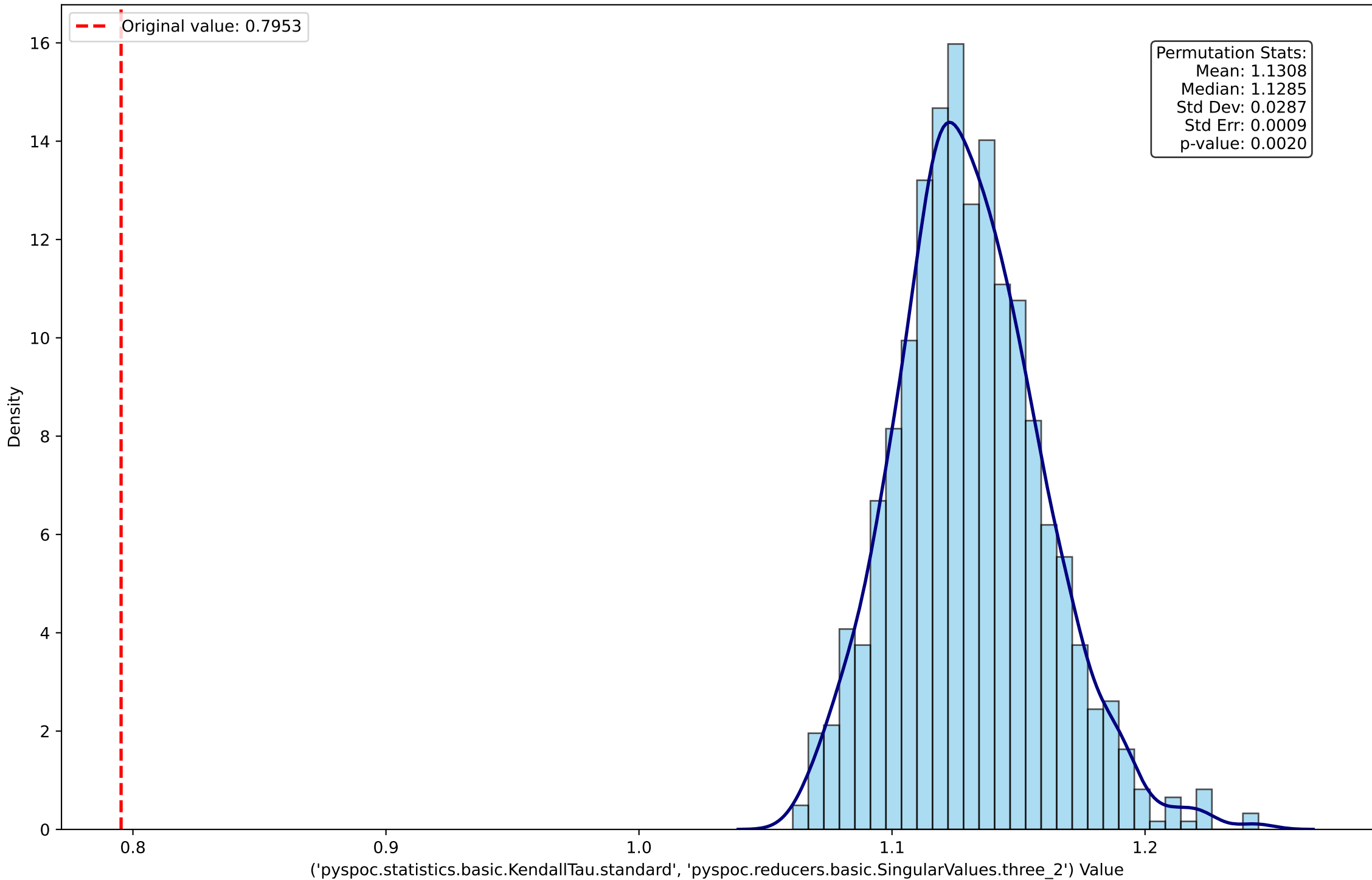
Permutation Distribution for ('pyspoc.statistics.basic.KendallTau.sq', 'pyspoc.reducers.basic.SingularValues.three_3')



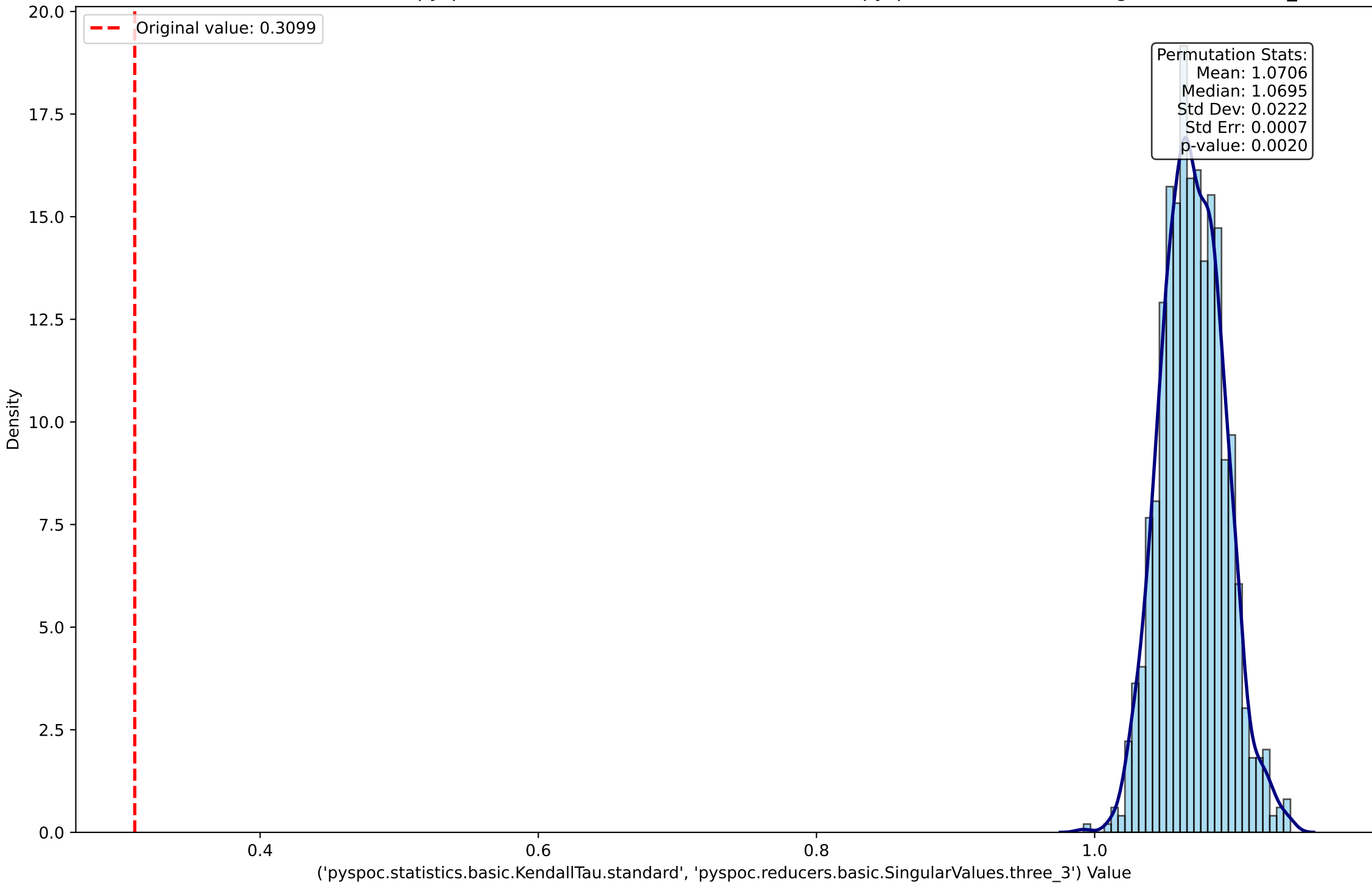
Permutation Distribution for ('pyspoc.statistics.basic.KendallTau.standard', 'pyspoc.reducers.basic.SingularValues.three_1')



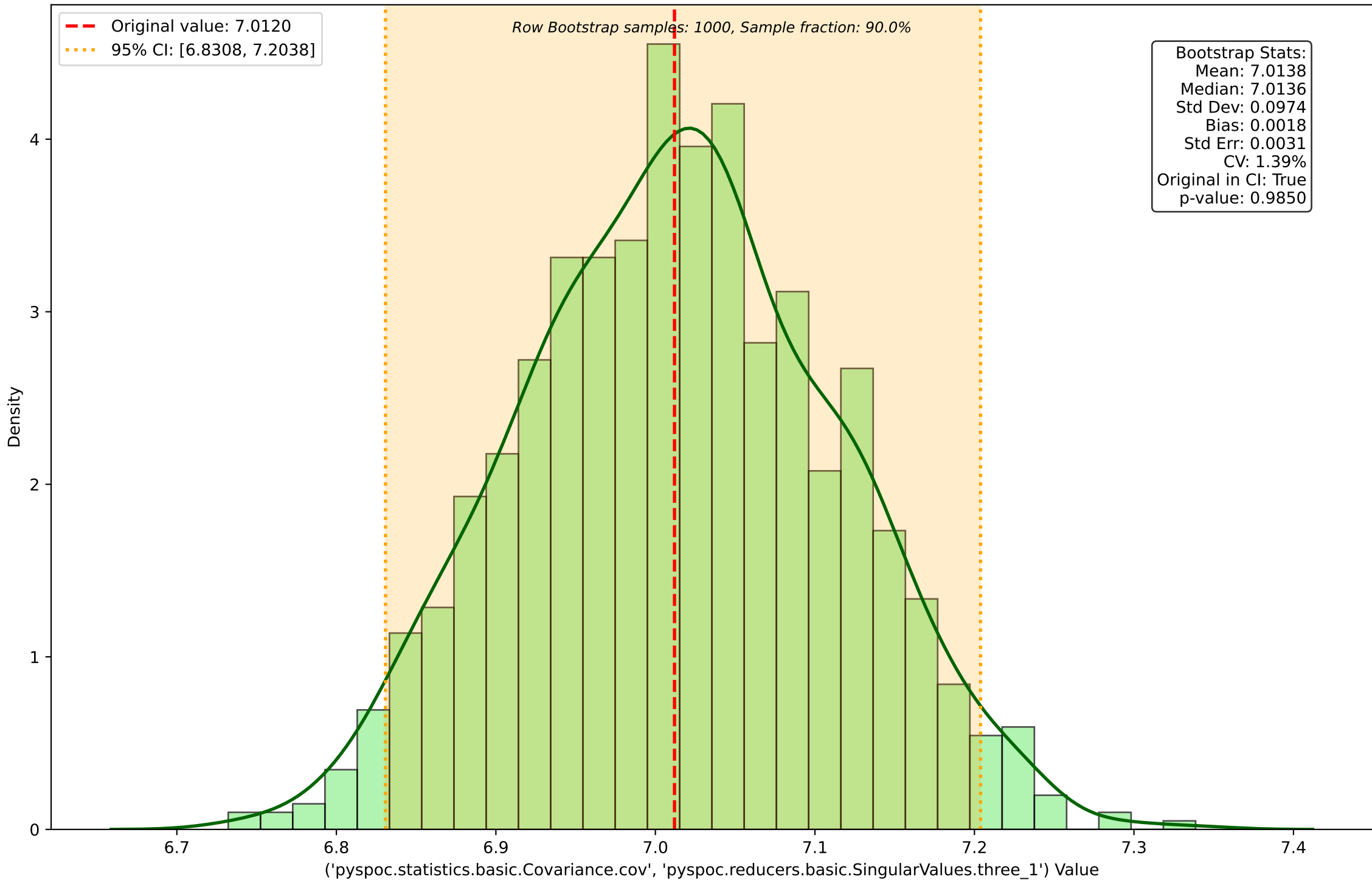
Permutation Distribution for ('pypoc.statistics.basic.KendallTau.standard', 'pypoc.reducers.basic.SingularValues.three_2')



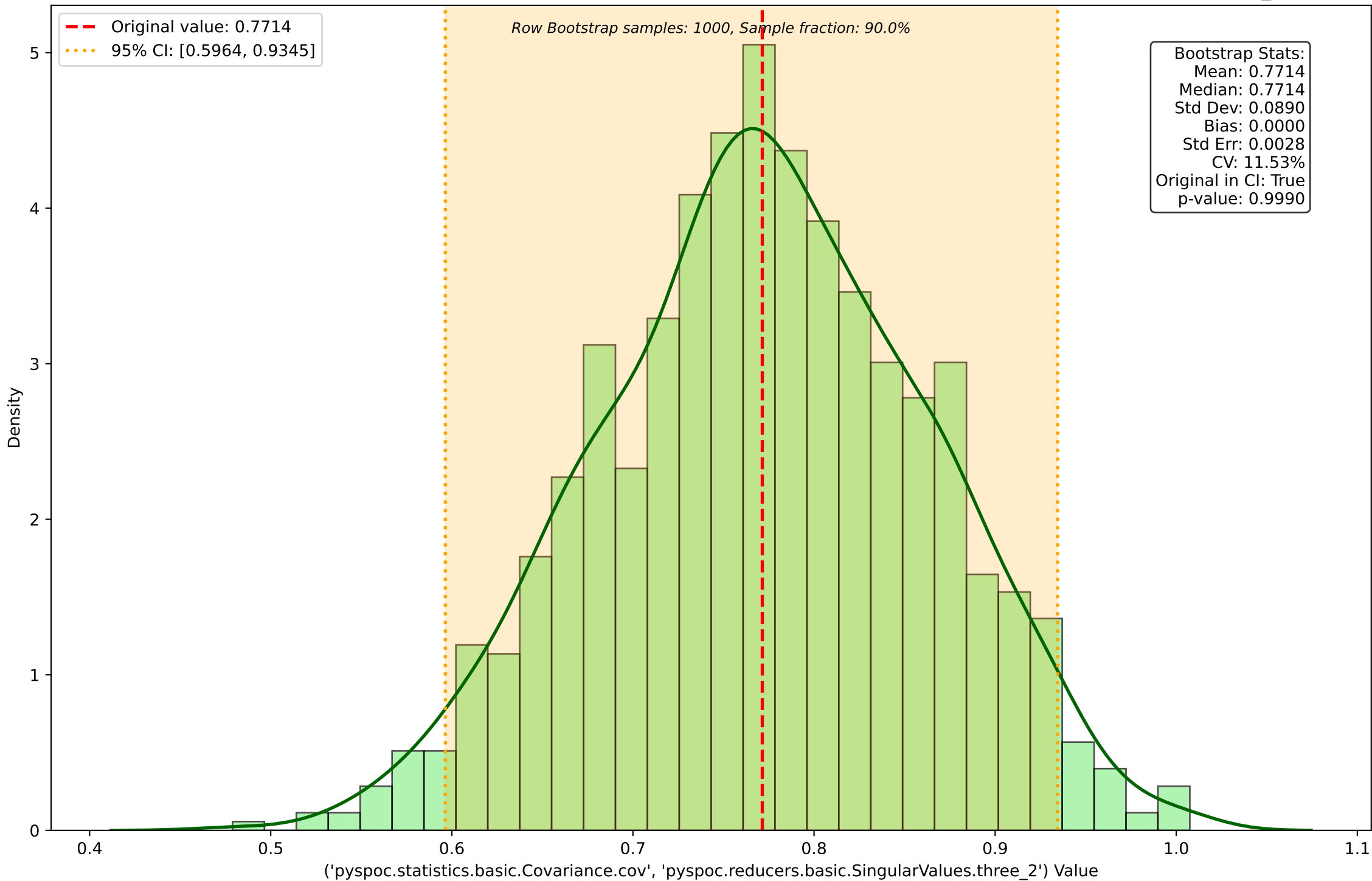
Permutation Distribution for ('pyspoc.statistics.basic.KendallTau.standard', 'pyspoc.reducers.basic.SingularValues.three_3')



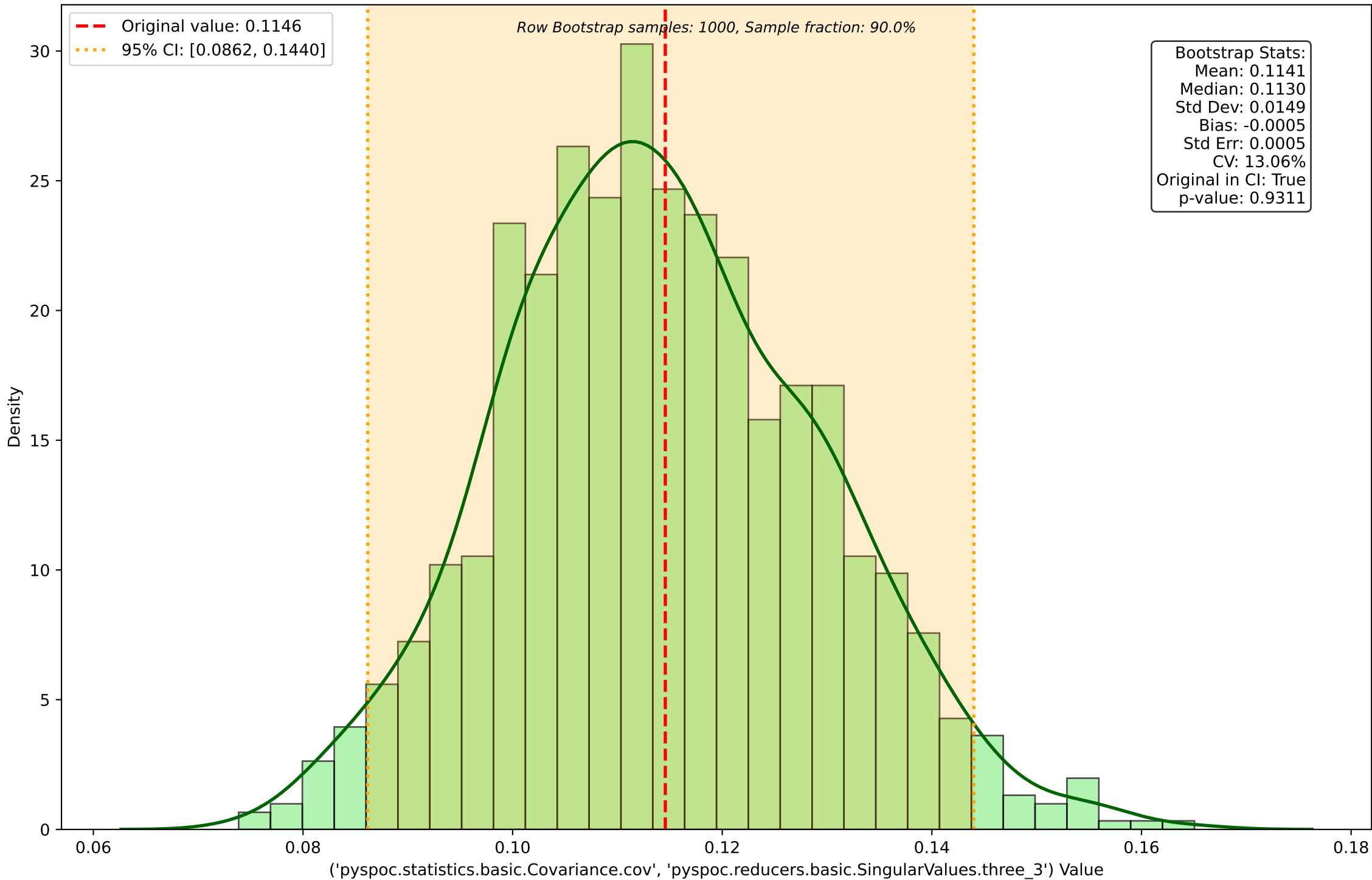
Row Bootstrap Distribution for ('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.SingularValues.three_1')



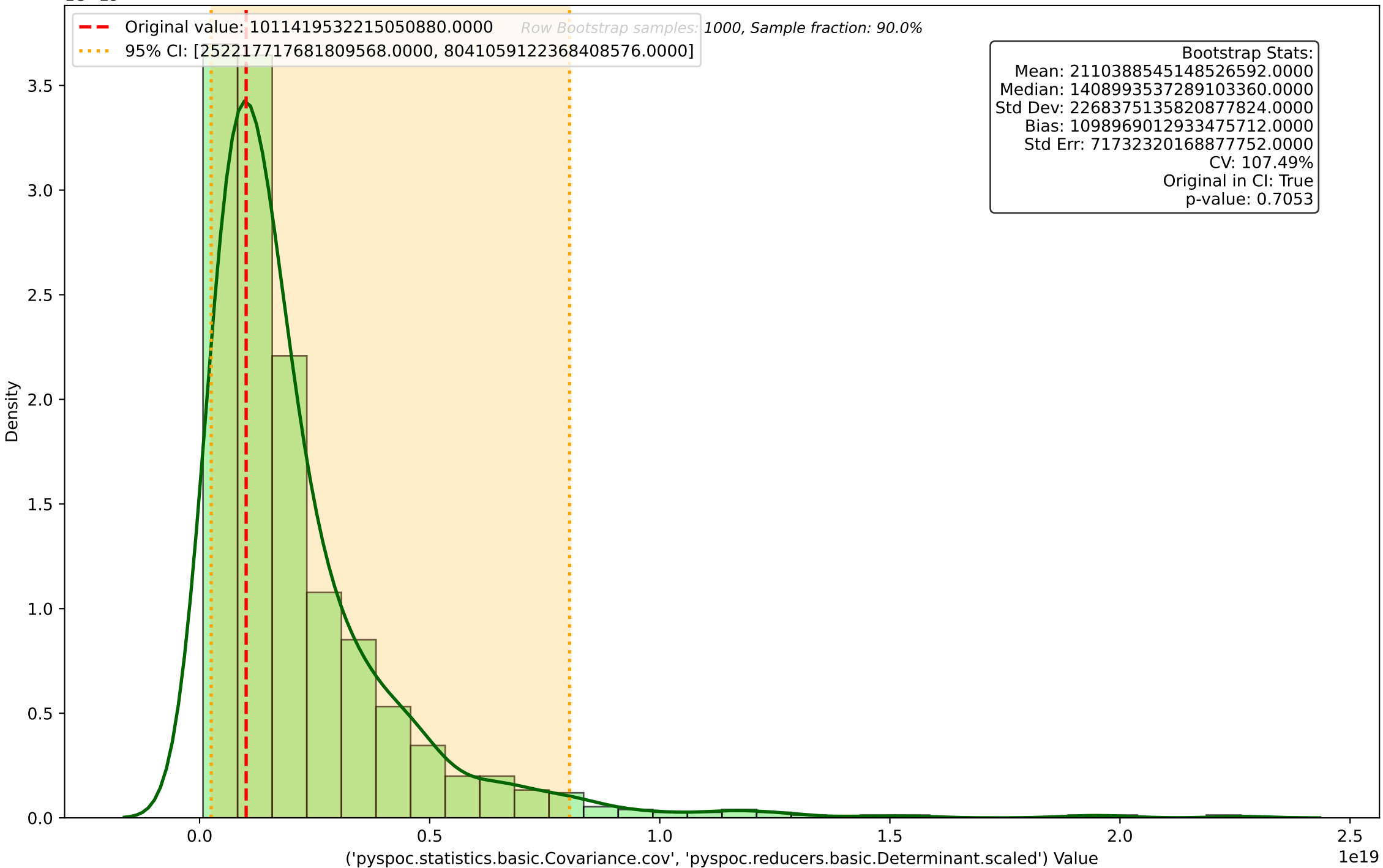
Row Bootstrap Distribution for ('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.SingularValues.three_2')



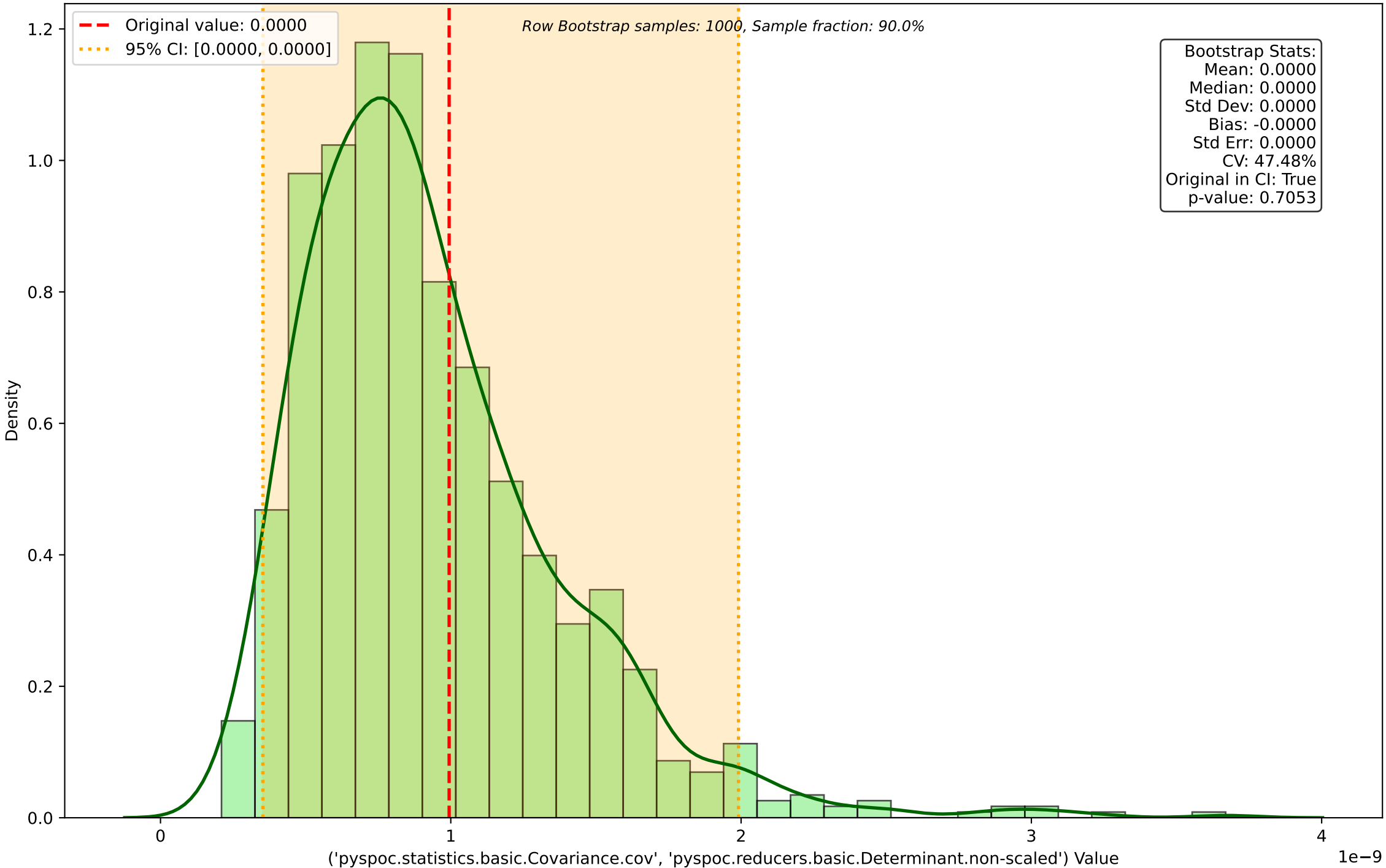
Row Bootstrap Distribution for ('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.SingularValues.three_3')



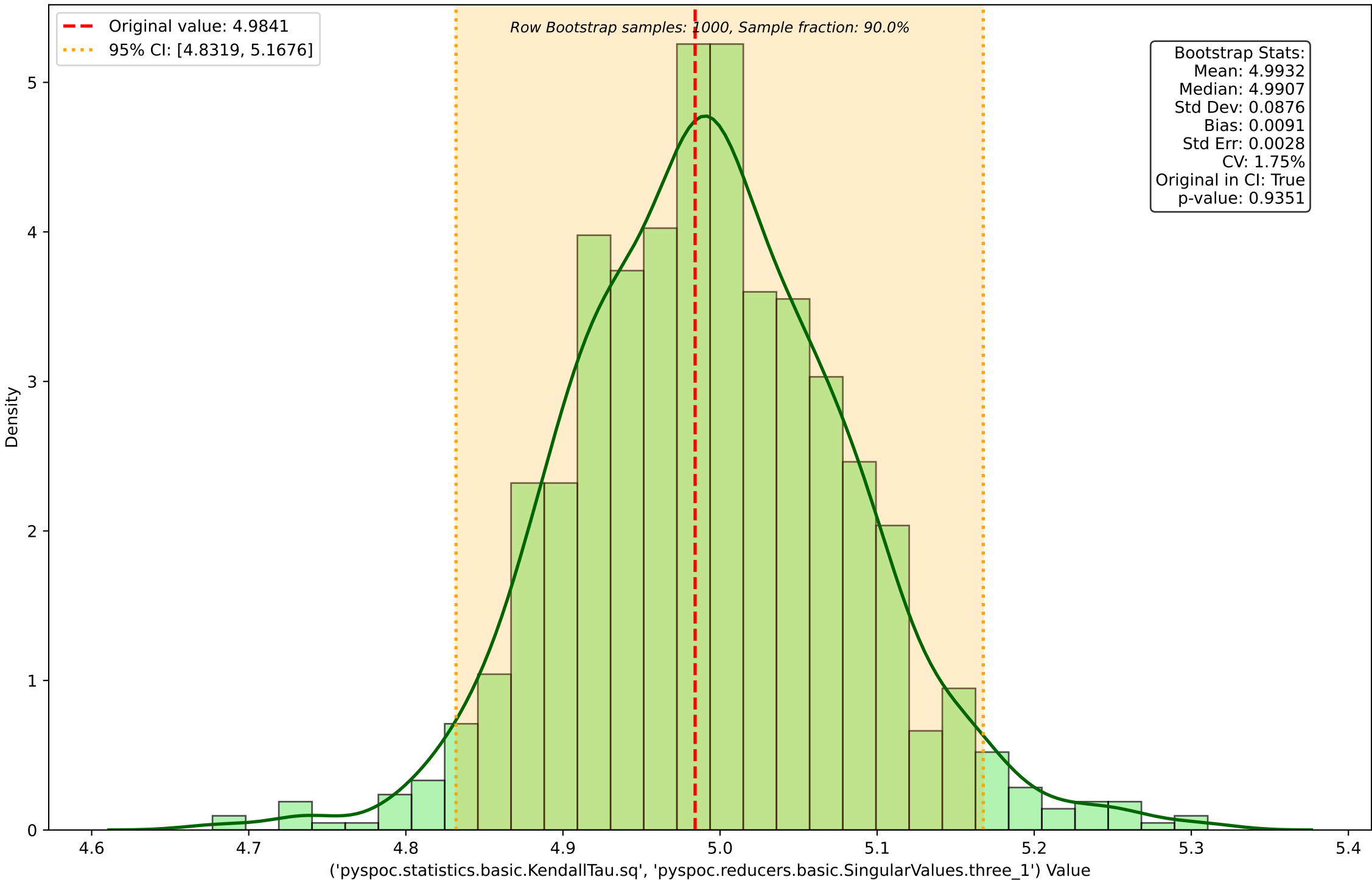
Row Bootstrap Distribution for ('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.Determinant.scaled')



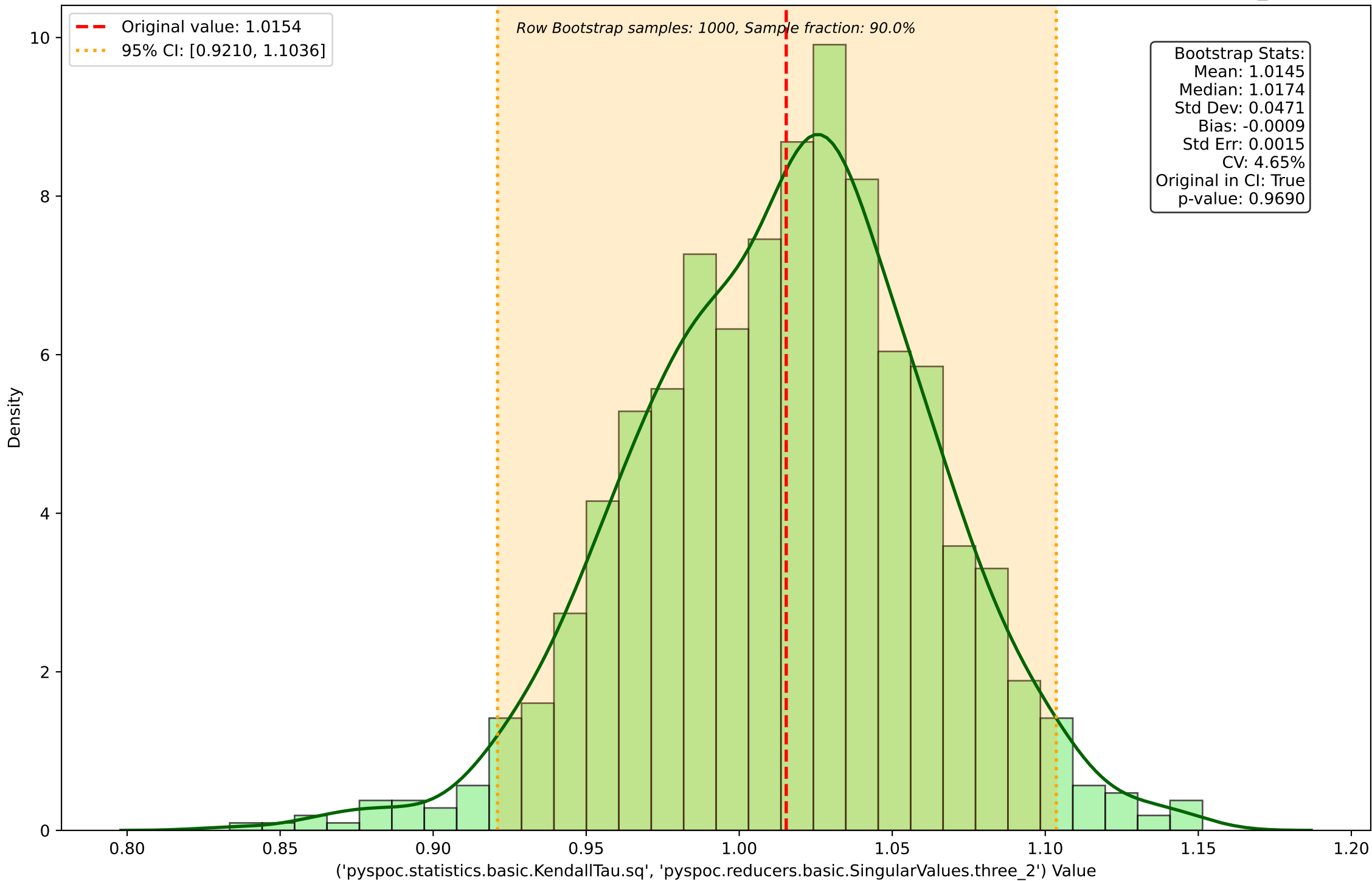
Row Bootstrap Distribution for ('pypoc.statistics.basic.Covariance.cov', 'pypoc.reducers.basic.Determinant.non-scaled')



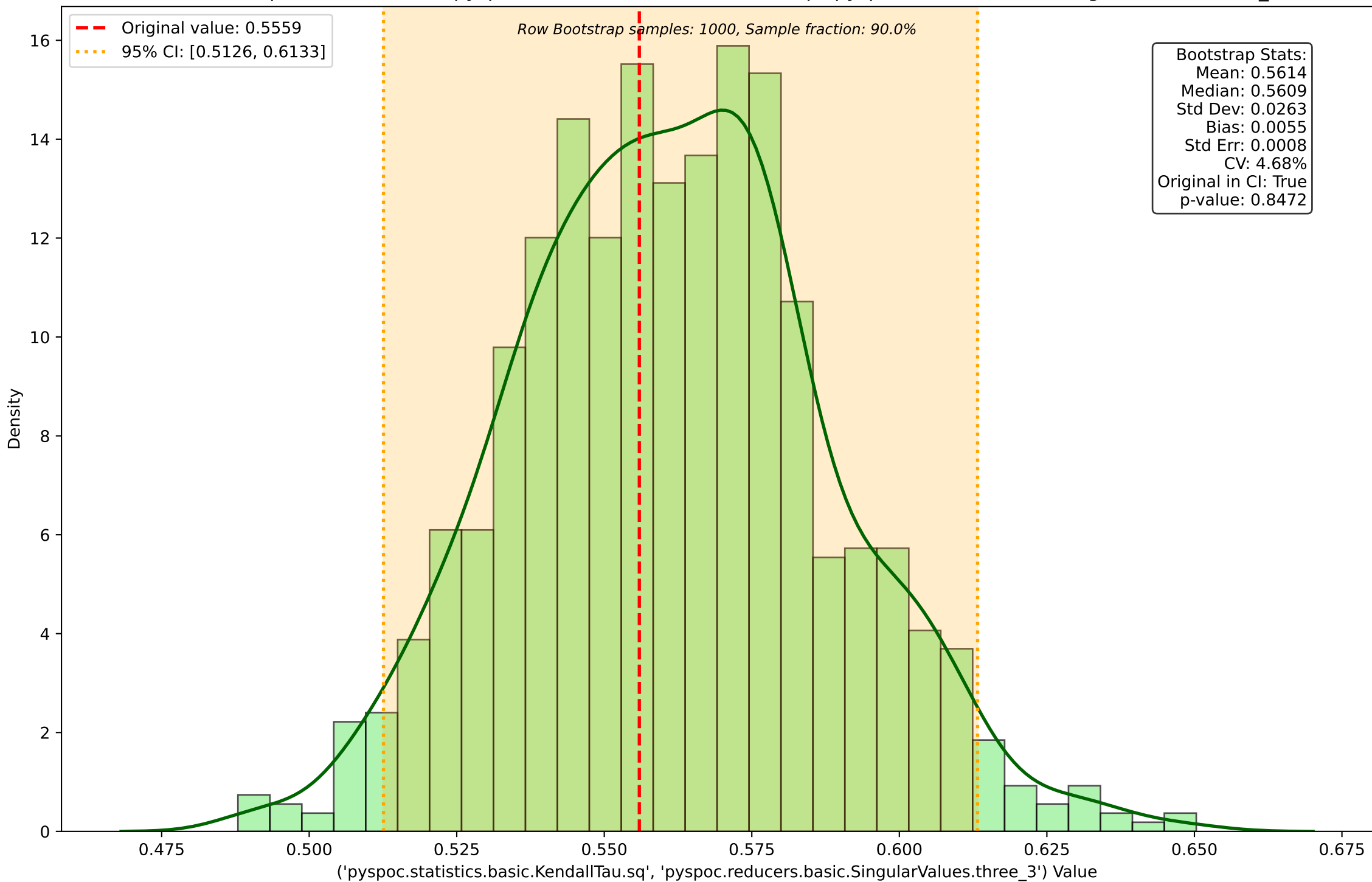
Row Bootstrap Distribution for ('pypoc.statistics.basic.KendallTau.sq', 'pypoc.reducers.basic.SingularValues.three_1')



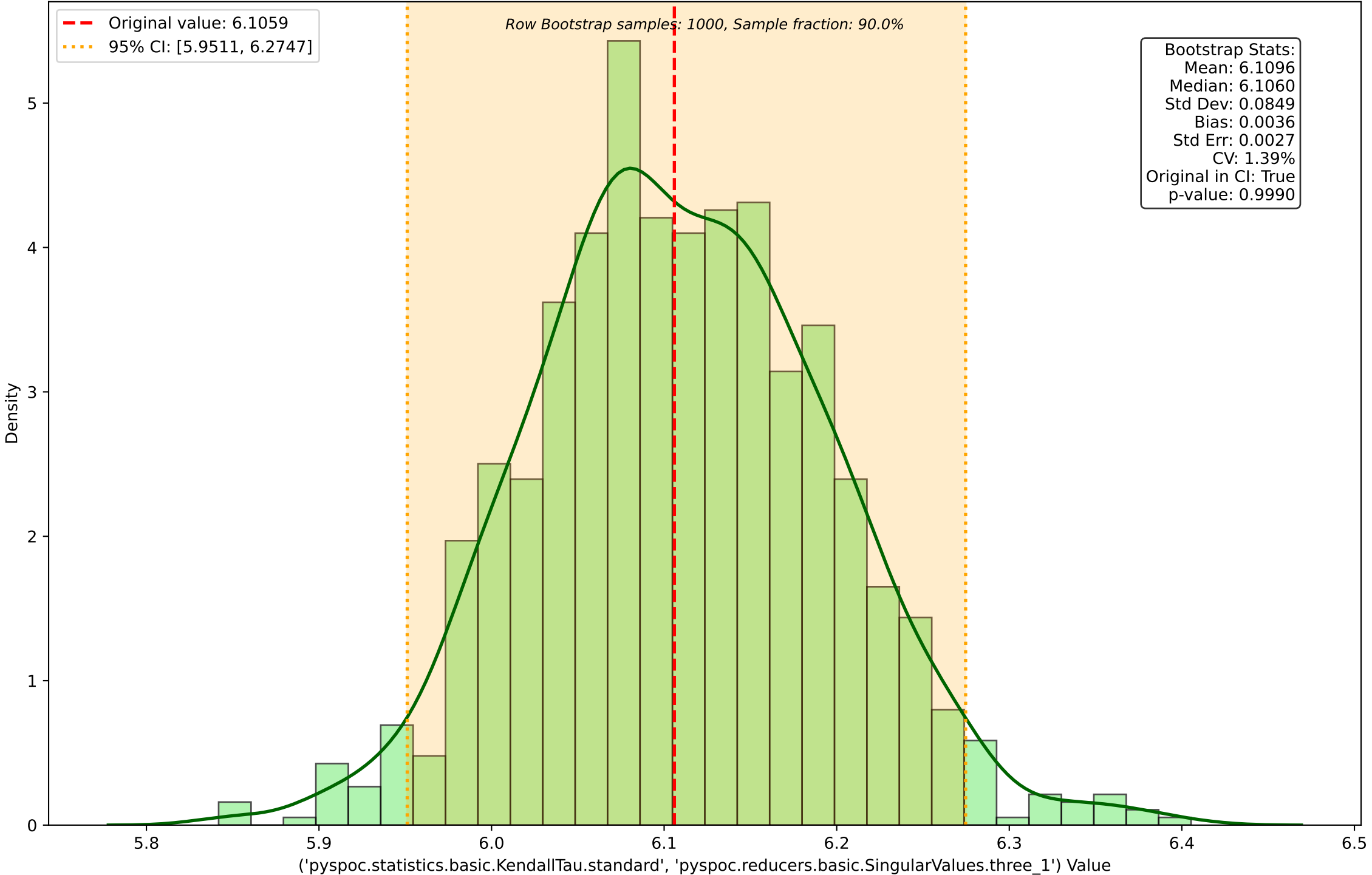
Row Bootstrap Distribution for ('pypoc.statistics.basic.KendallTau.sq', 'pypoc.reducers.basic.SingularValues.three_2')



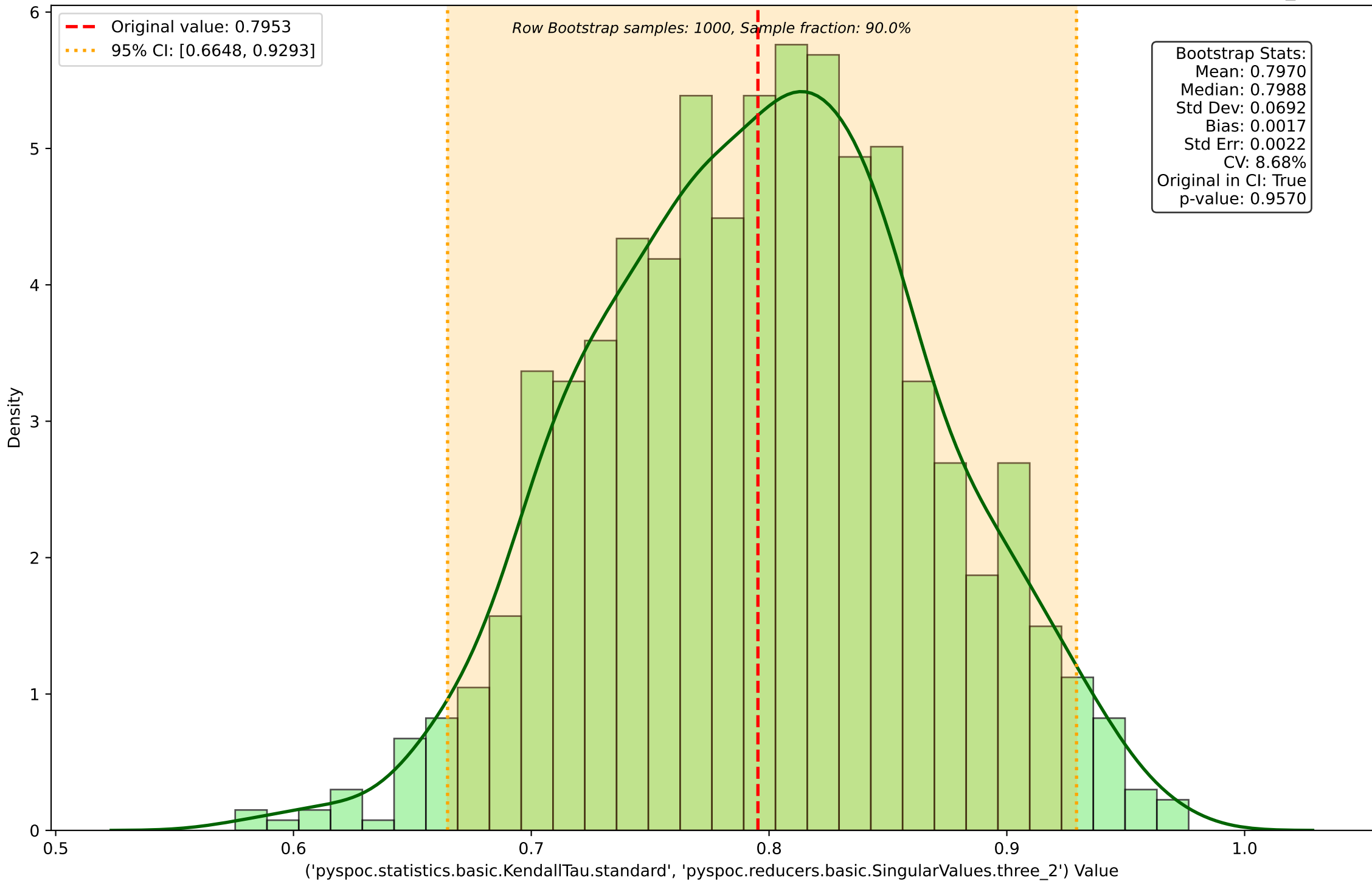
Row Bootstrap Distribution for ('pyspoc.statistics.basic.KendallTau.sq', 'pyspoc.reducers.basic.SingularValues.three_3')



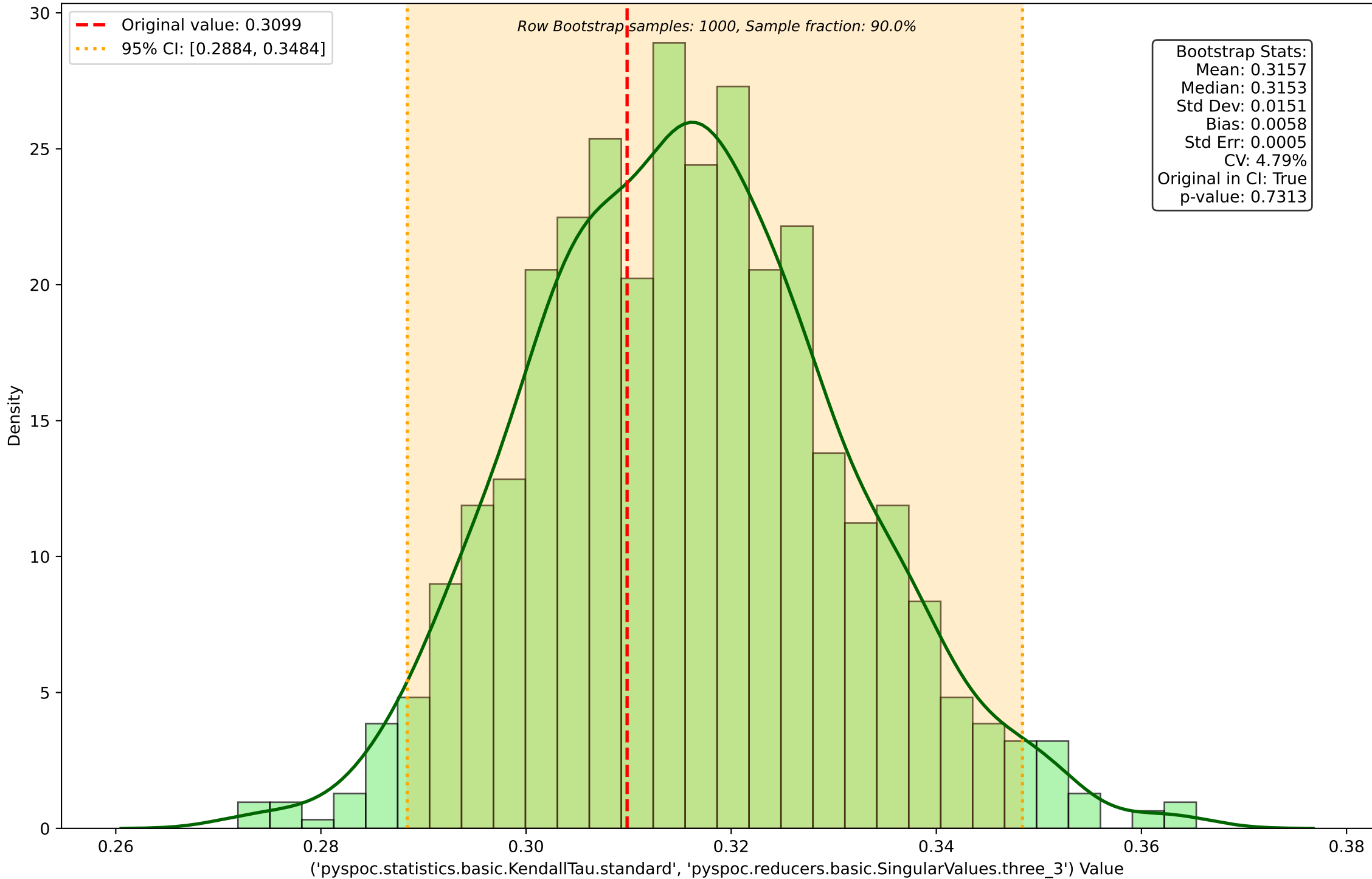
Row Bootstrap Distribution for ('pyspoc.statistics.basic.KendallTau.standard', 'pyspoc.reducers.basic.SingularValues.three_1')



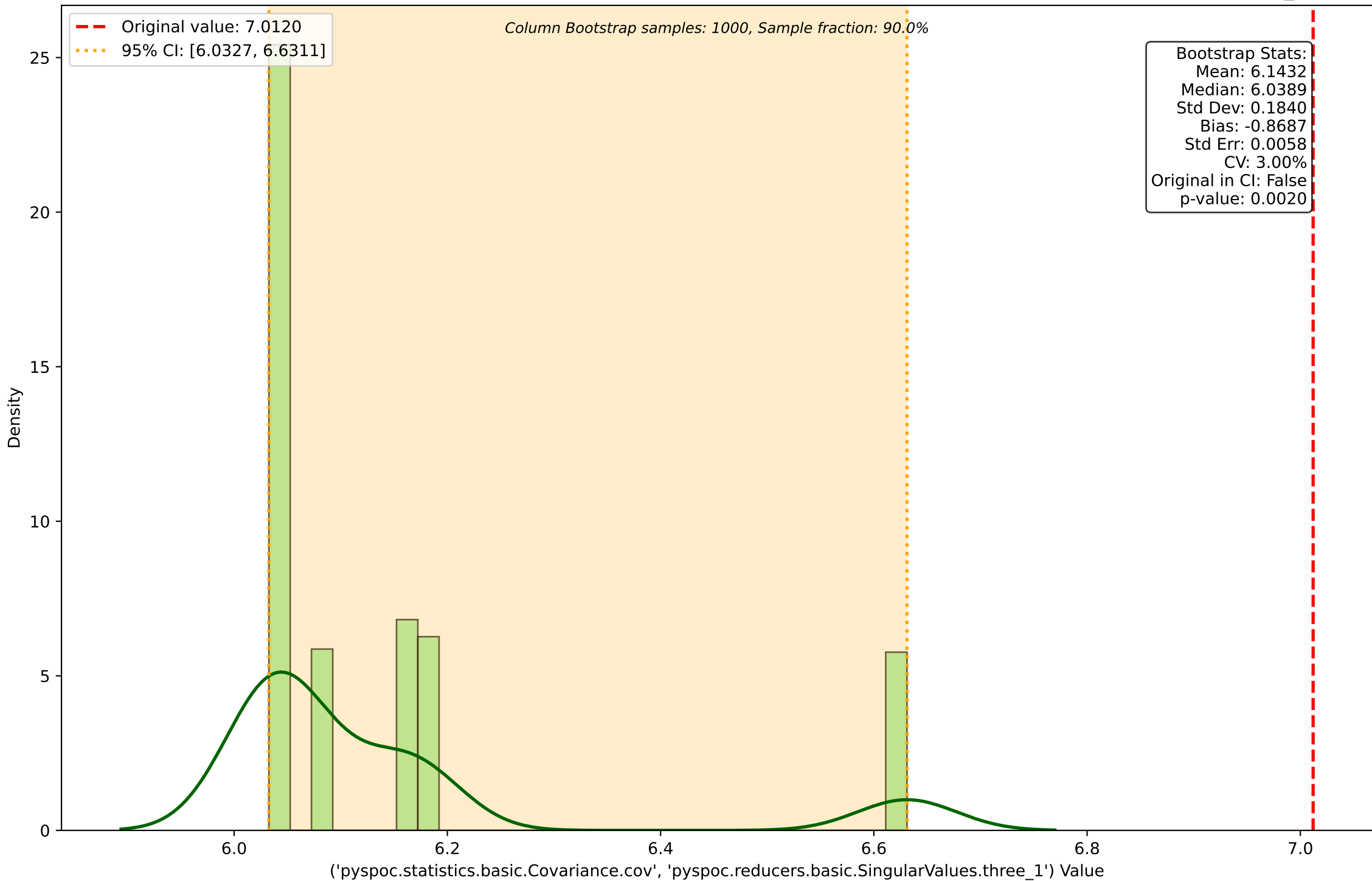
Row Bootstrap Distribution for ('pyspoc.statistics.basic.KendallTau.standard', 'pyspoc.reducers.basic.SingularValues.three_2')



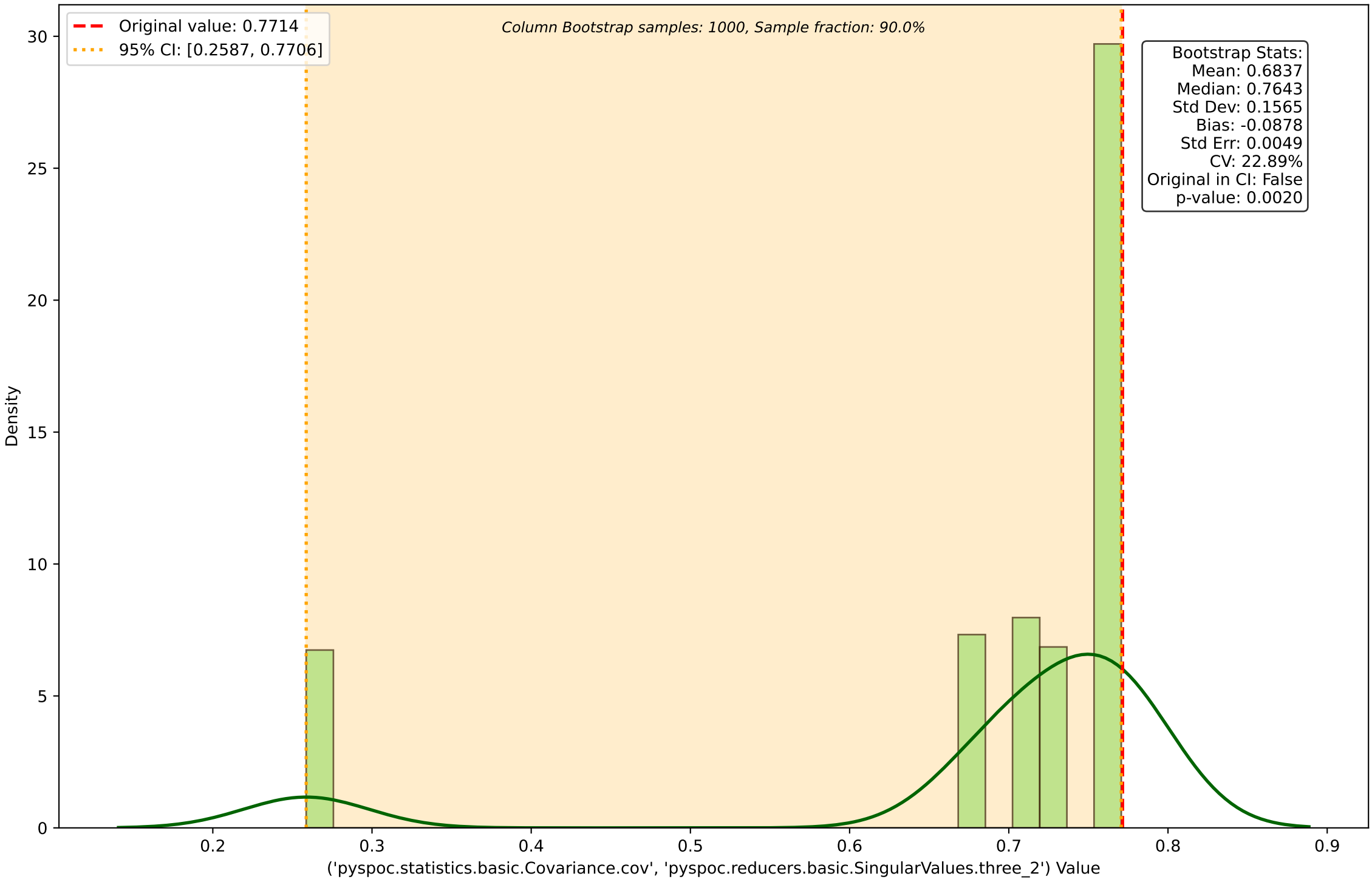
Row Bootstrap Distribution for ('pyspoc.statistics.basic.KendallTau.standard', 'pyspoc.reducers.basic.SingularValues.three_3')



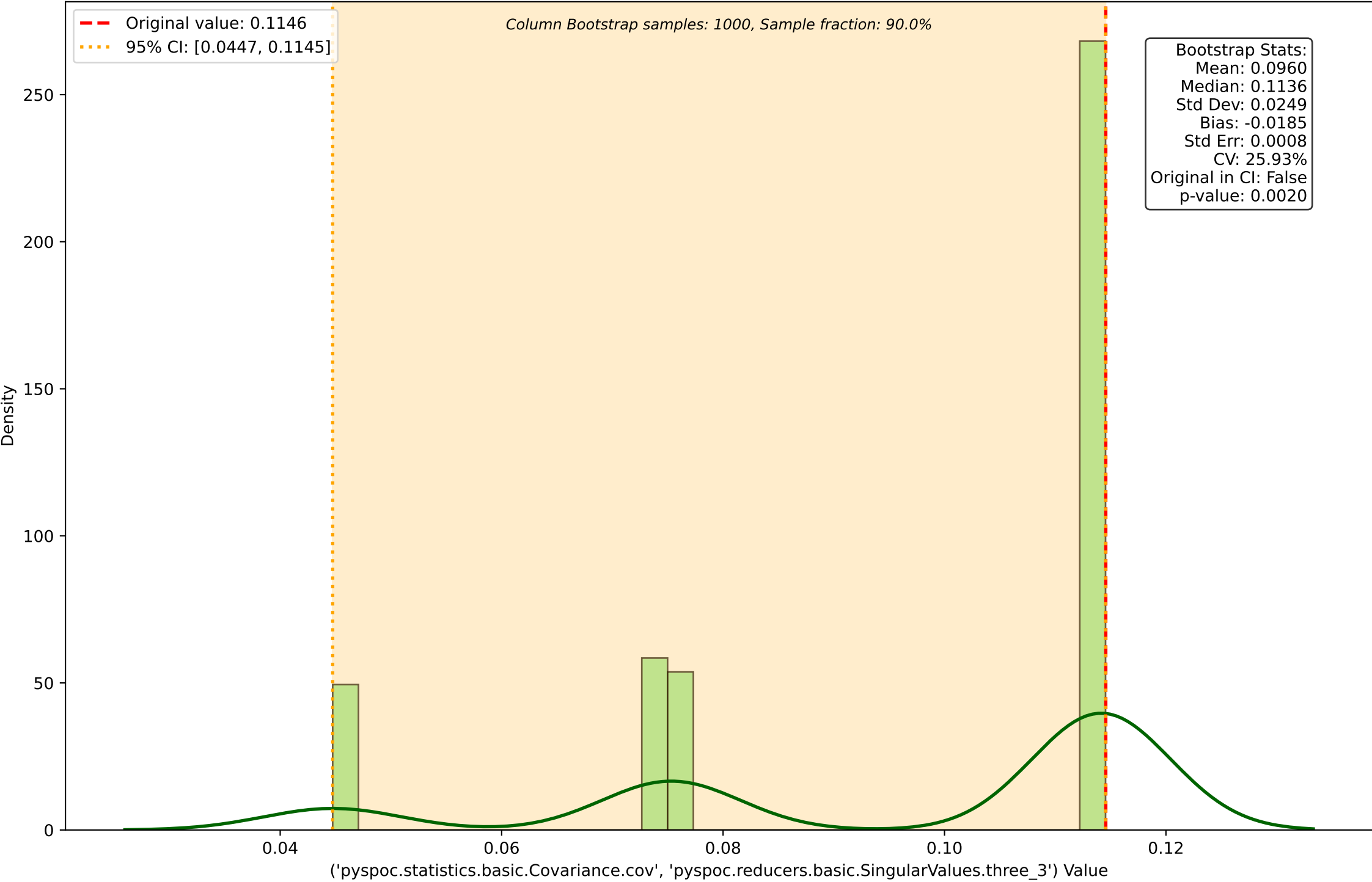
Column Bootstrap Distribution for ('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.SingularValues.three_1')

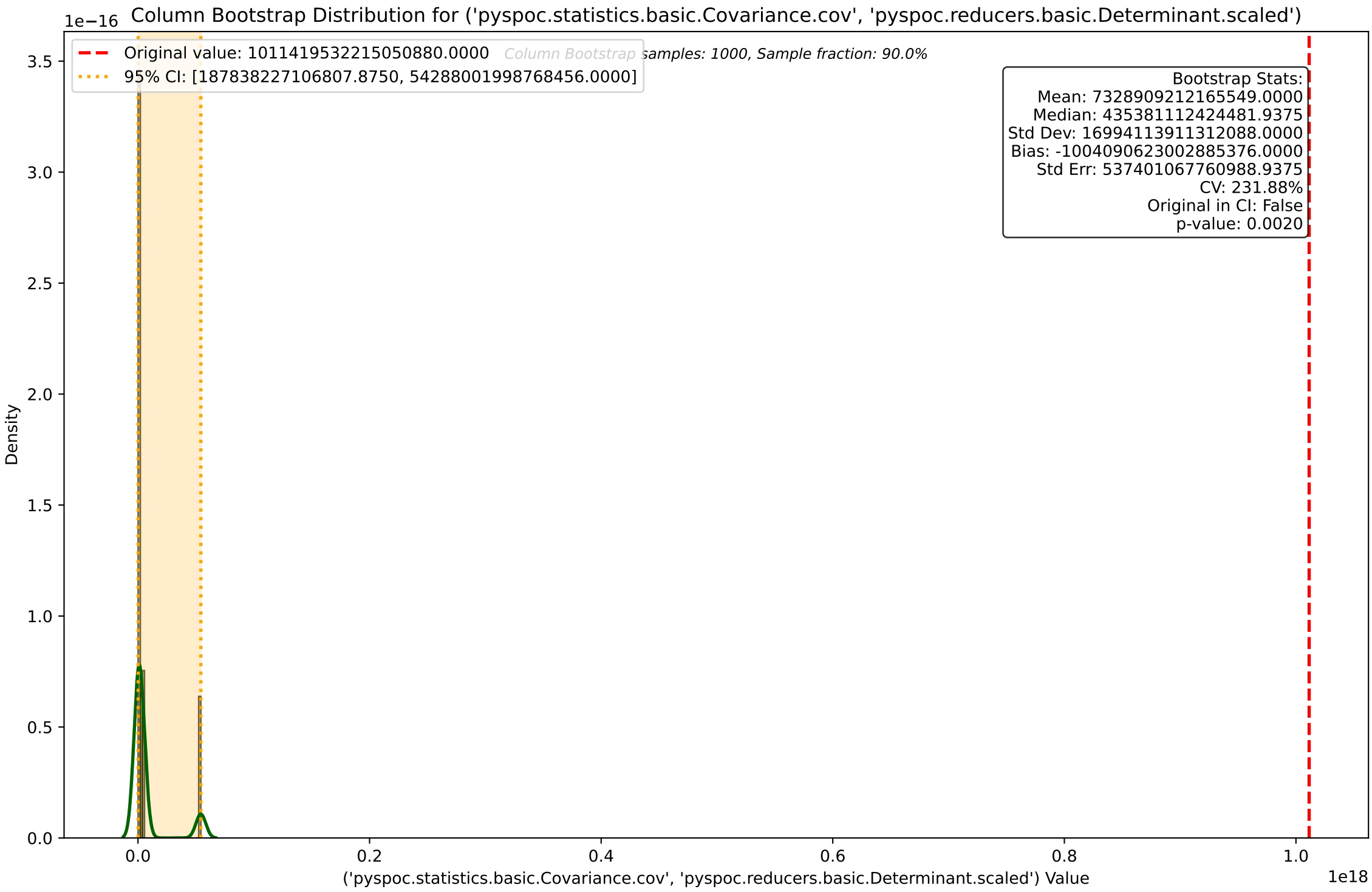


Column Bootstrap Distribution for ('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.SingularValues.three_2')

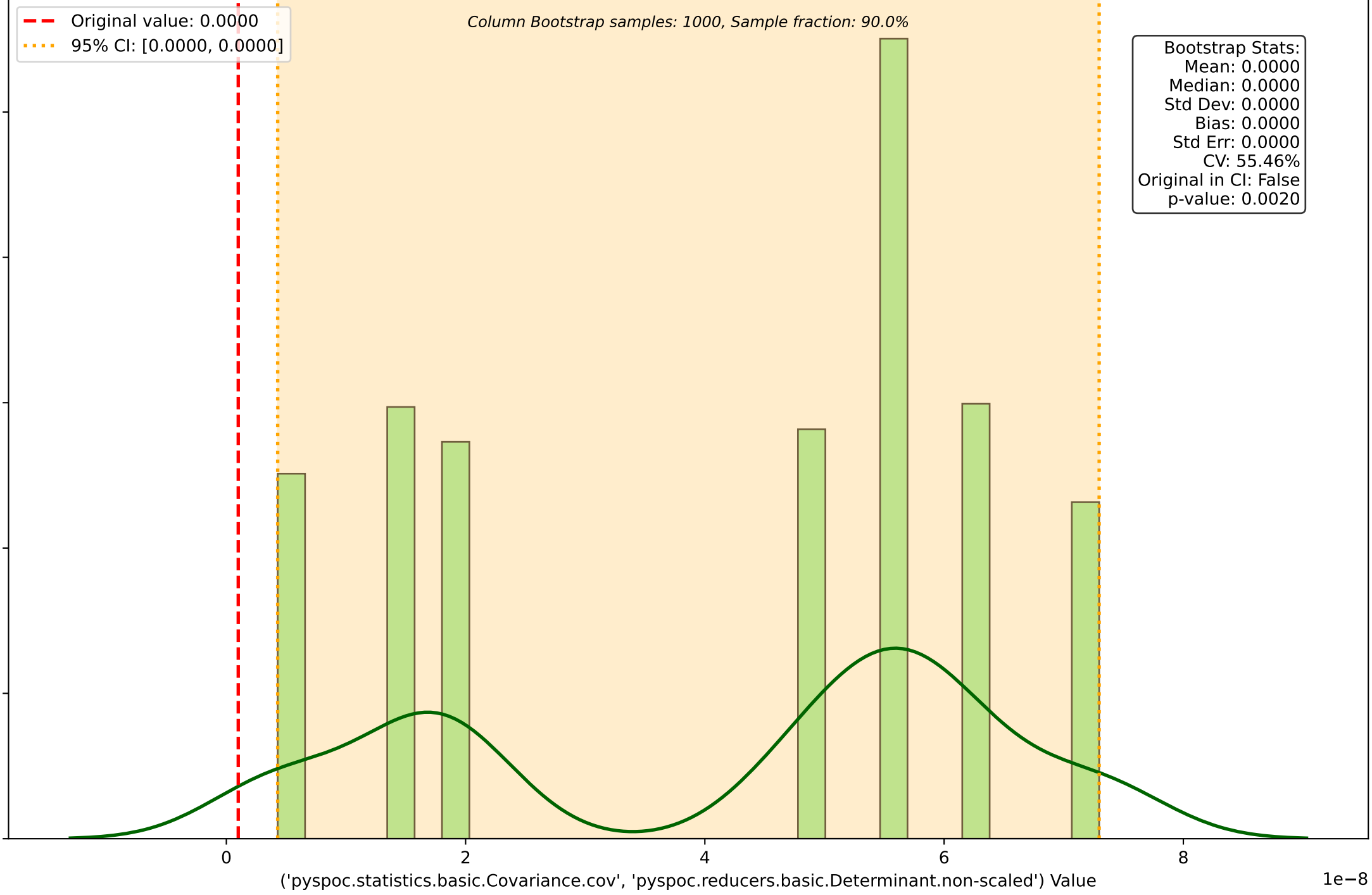


Column Bootstrap Distribution for ('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.SingularValues.three_3')

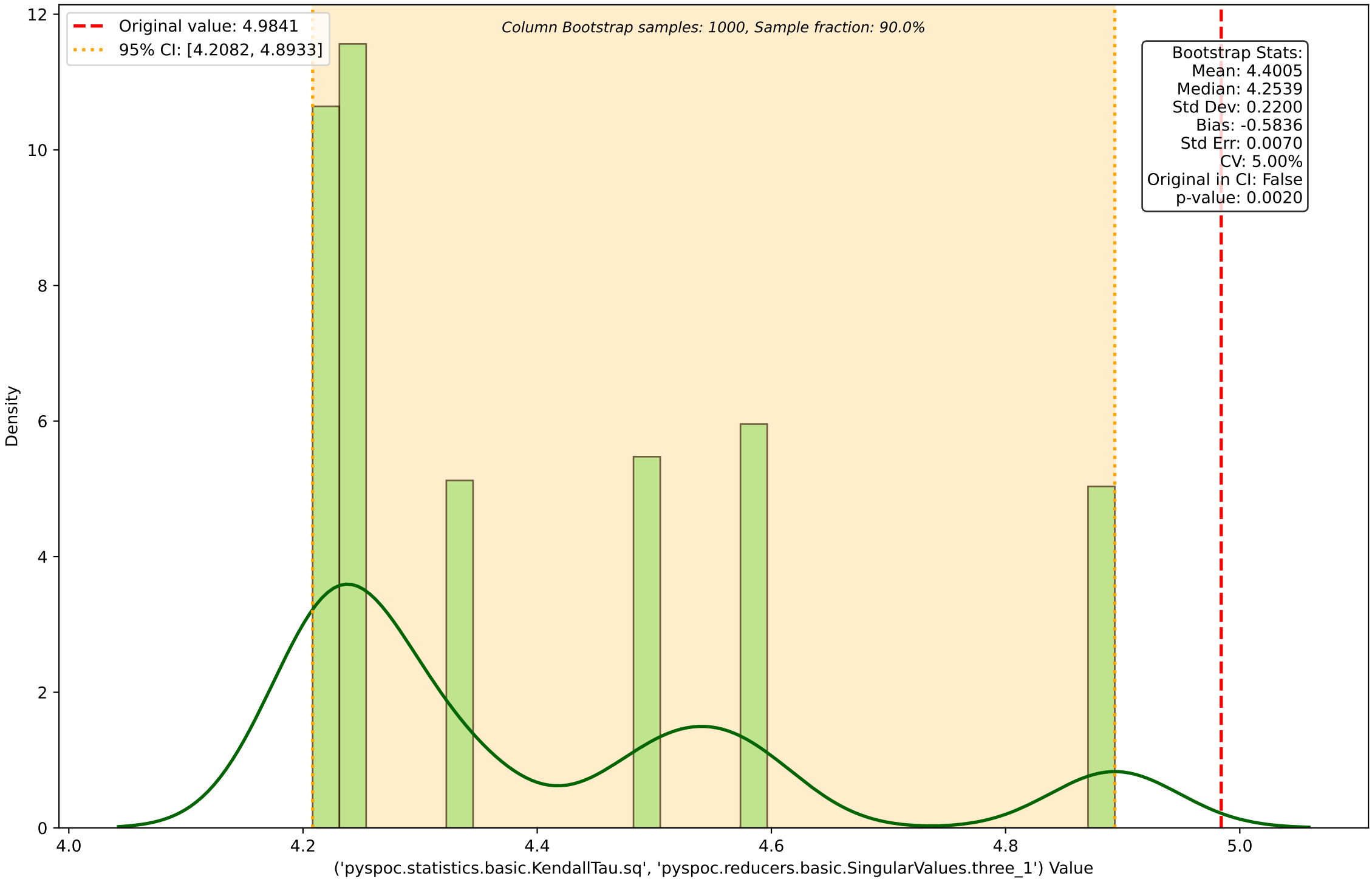




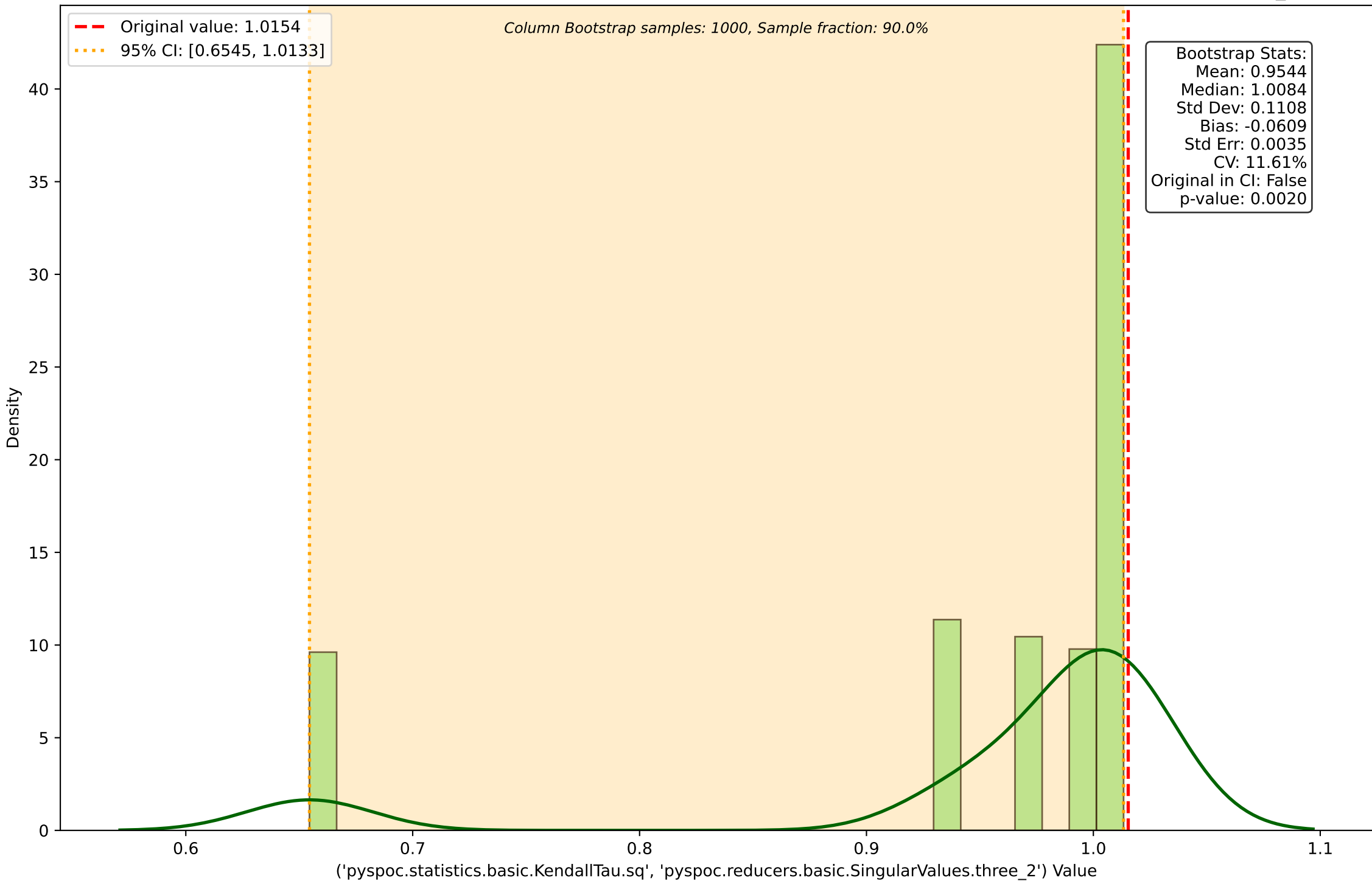
1e8 Column Bootstrap Distribution for ('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.Determinant.non-scaled')



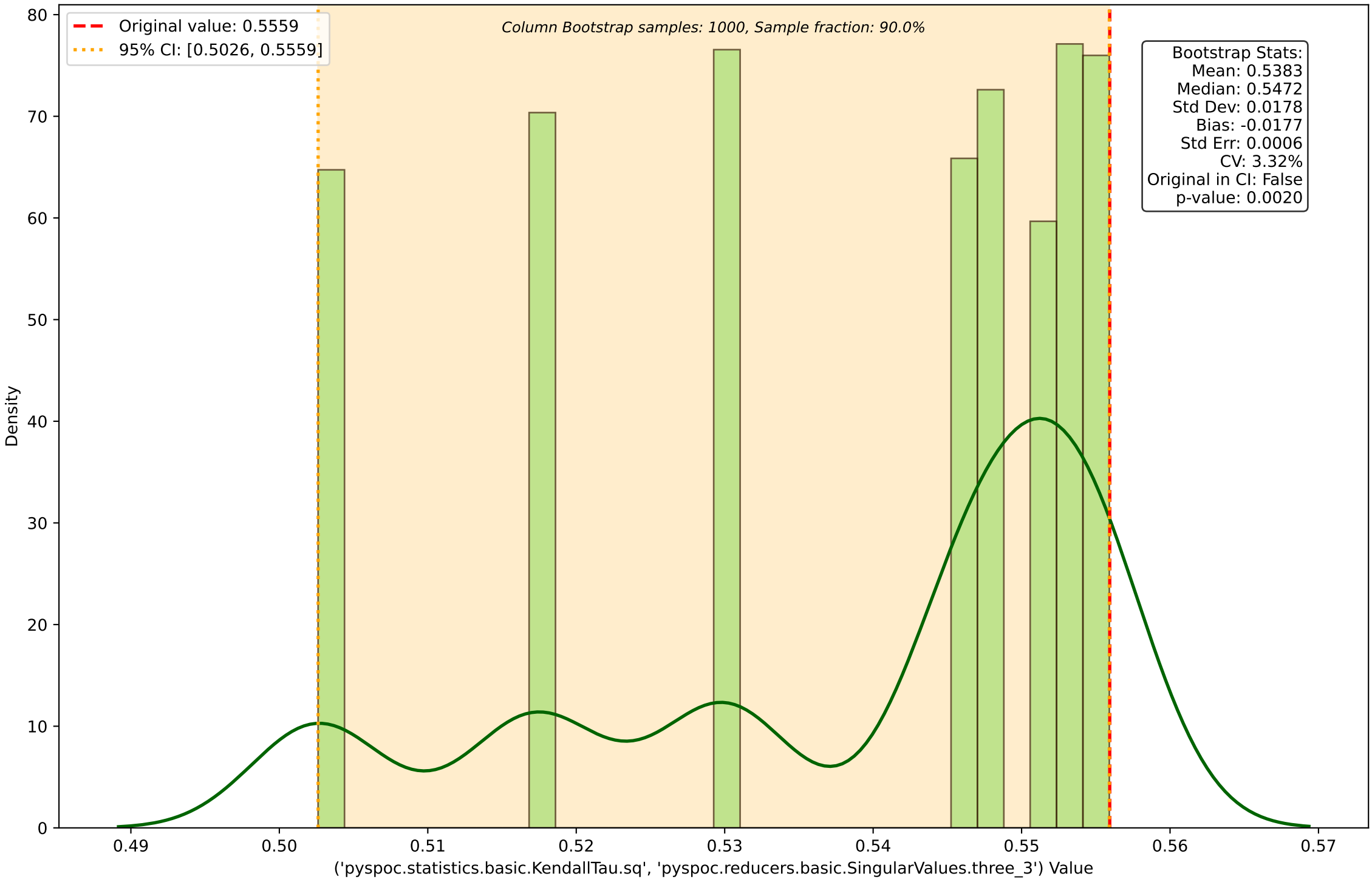
Column Bootstrap Distribution for ('pyspoc.statistics.basic.KendallTau.sq', 'pyspoc.reducers.basic.SingularValues.three_1')



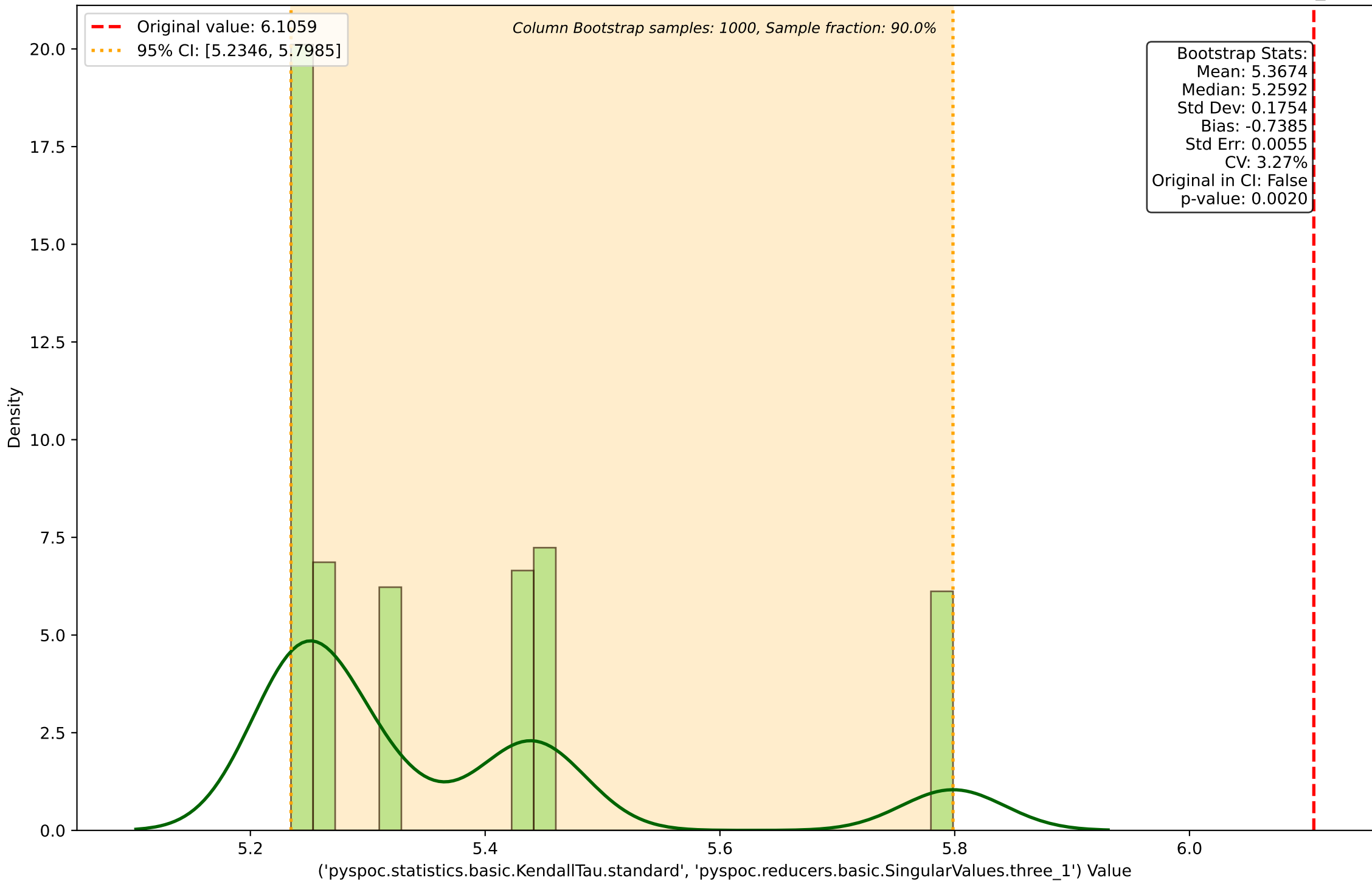
Column Bootstrap Distribution for ('pyspoc.statistics.basic.KendallTau.sq', 'pyspoc.reducers.basic.SingularValues.three_2')



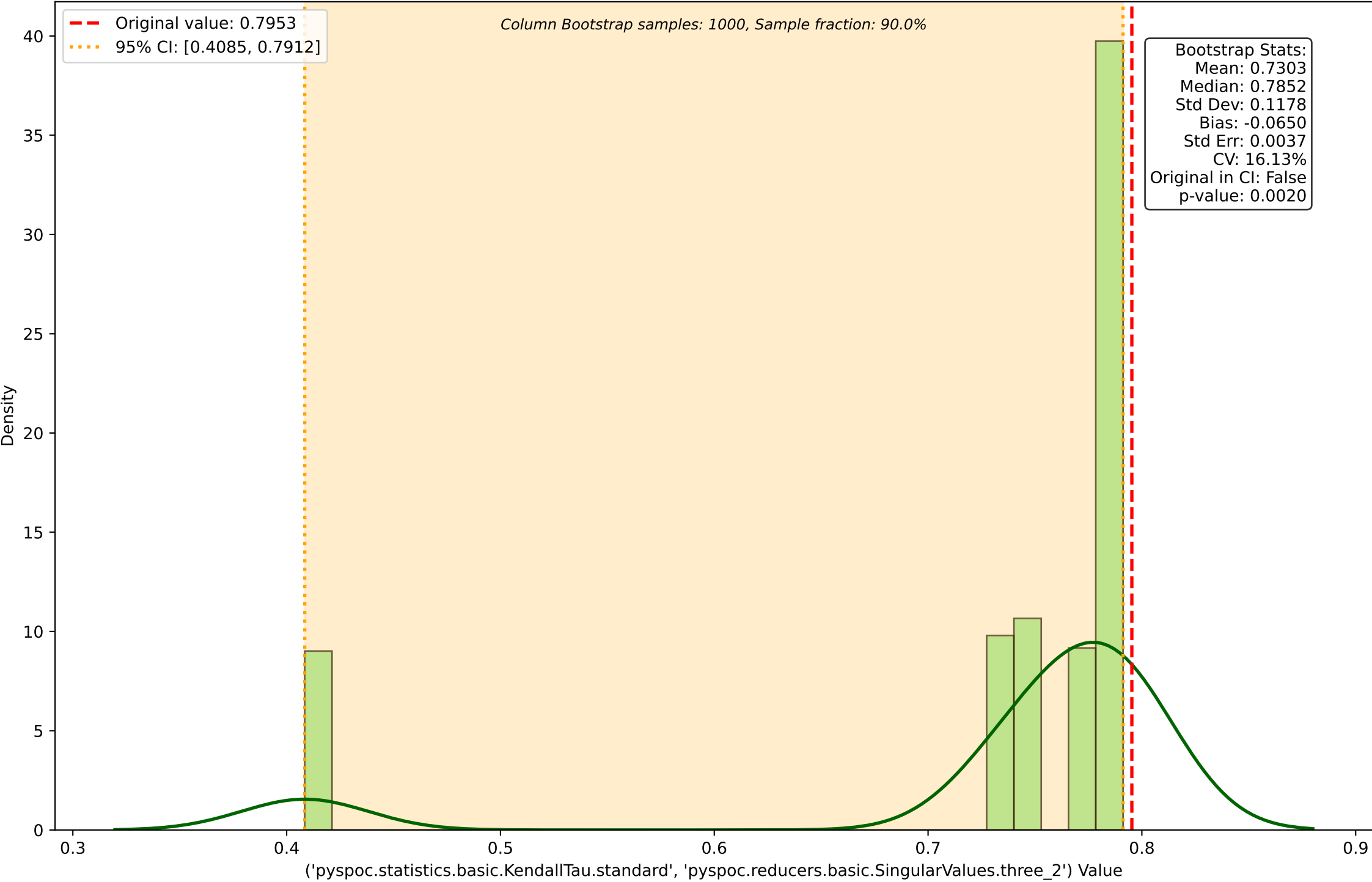
Column Bootstrap Distribution for ('pyspoc.statistics.basic.KendallTau.sq', 'pyspoc.reducers.basic.SingularValues.three_3')



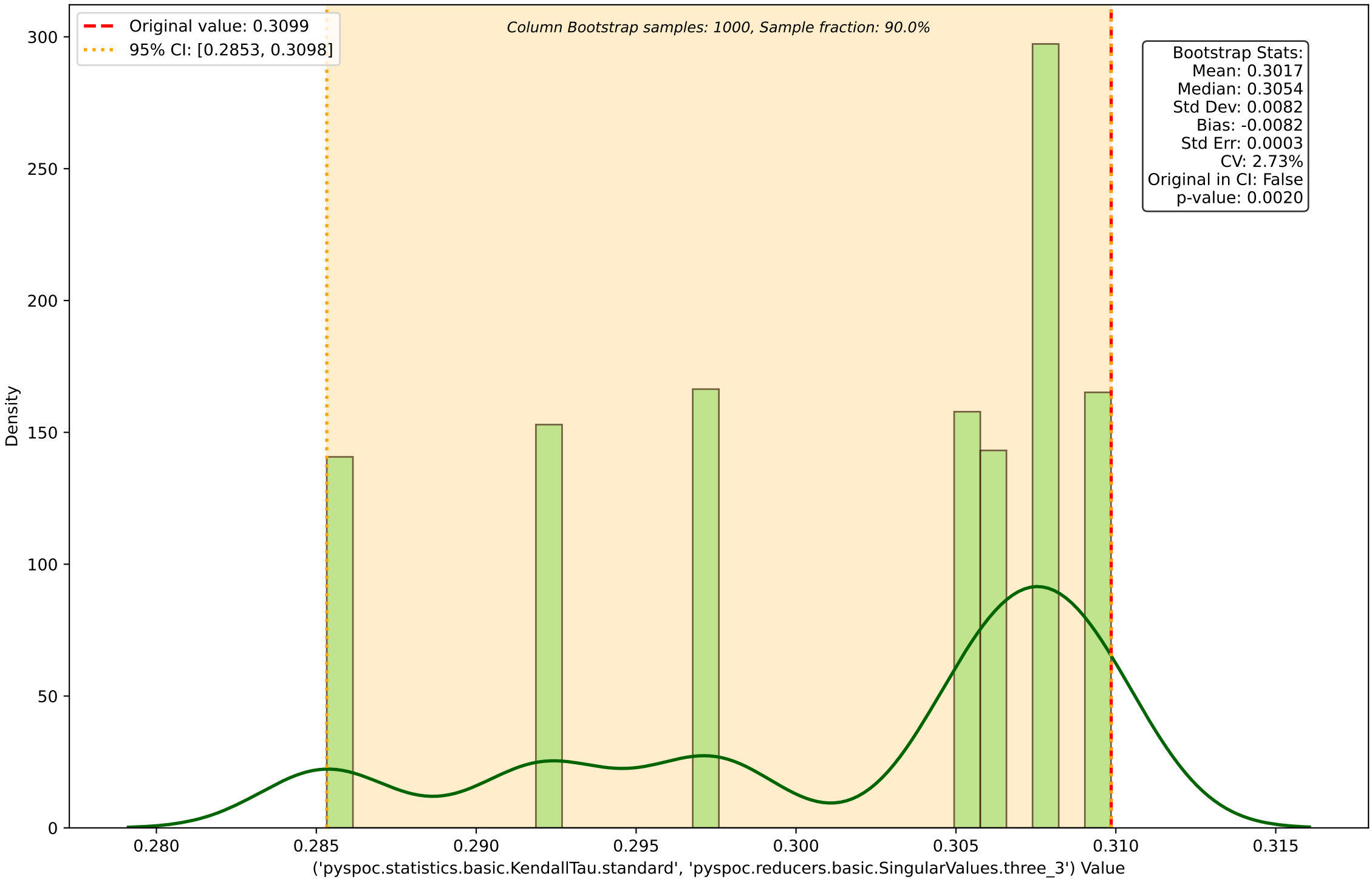
Column Bootstrap Distribution for ('pyspoc.statistics.basic.KendallTau.standard', 'pyspoc.reducers.basic.SingularValues.three_1')



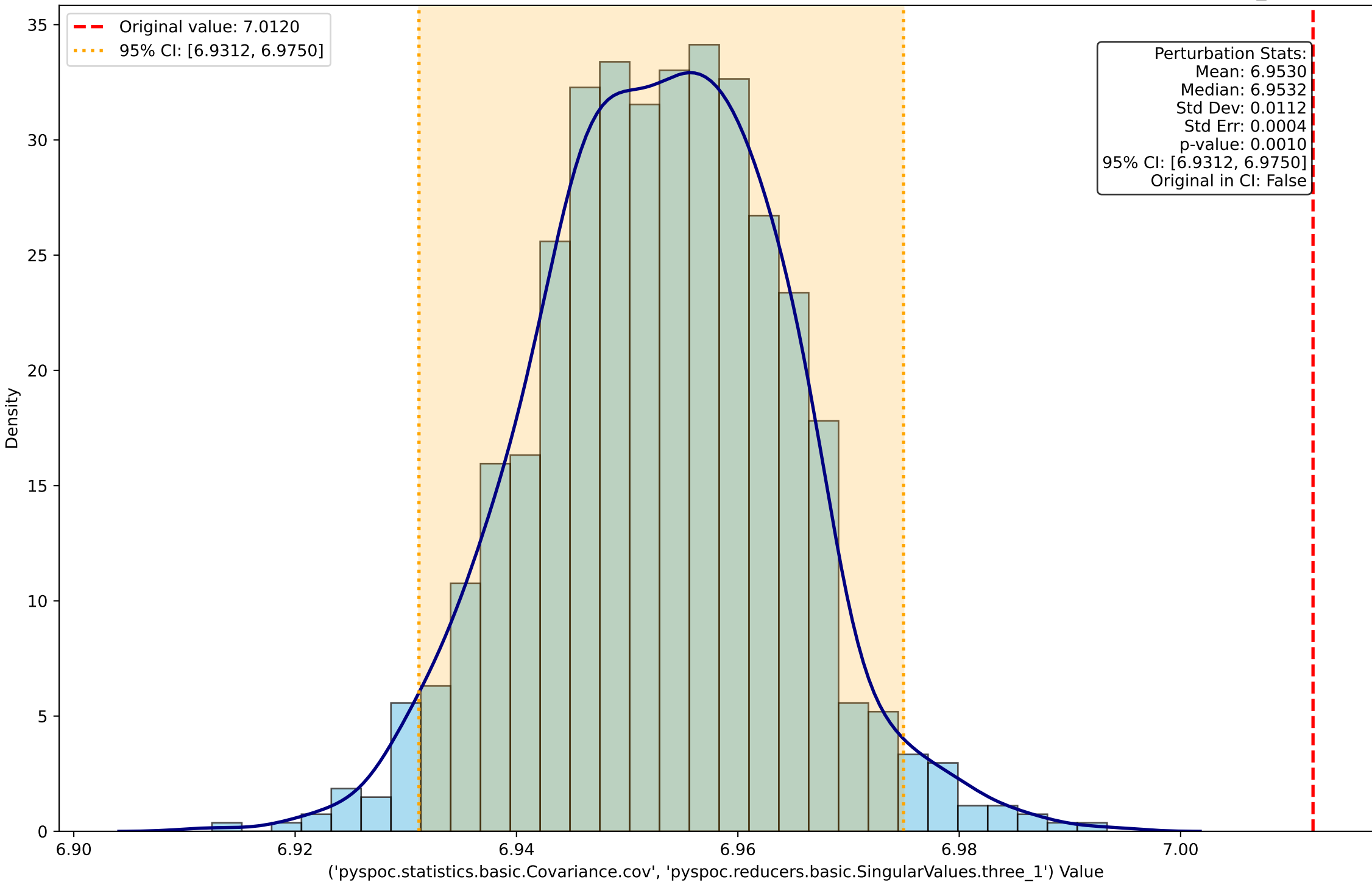
Column Bootstrap Distribution for ('pypoc.statistics.basic.KendallTau.standard', 'pypoc.reducers.basic.SingularValues.three_2')



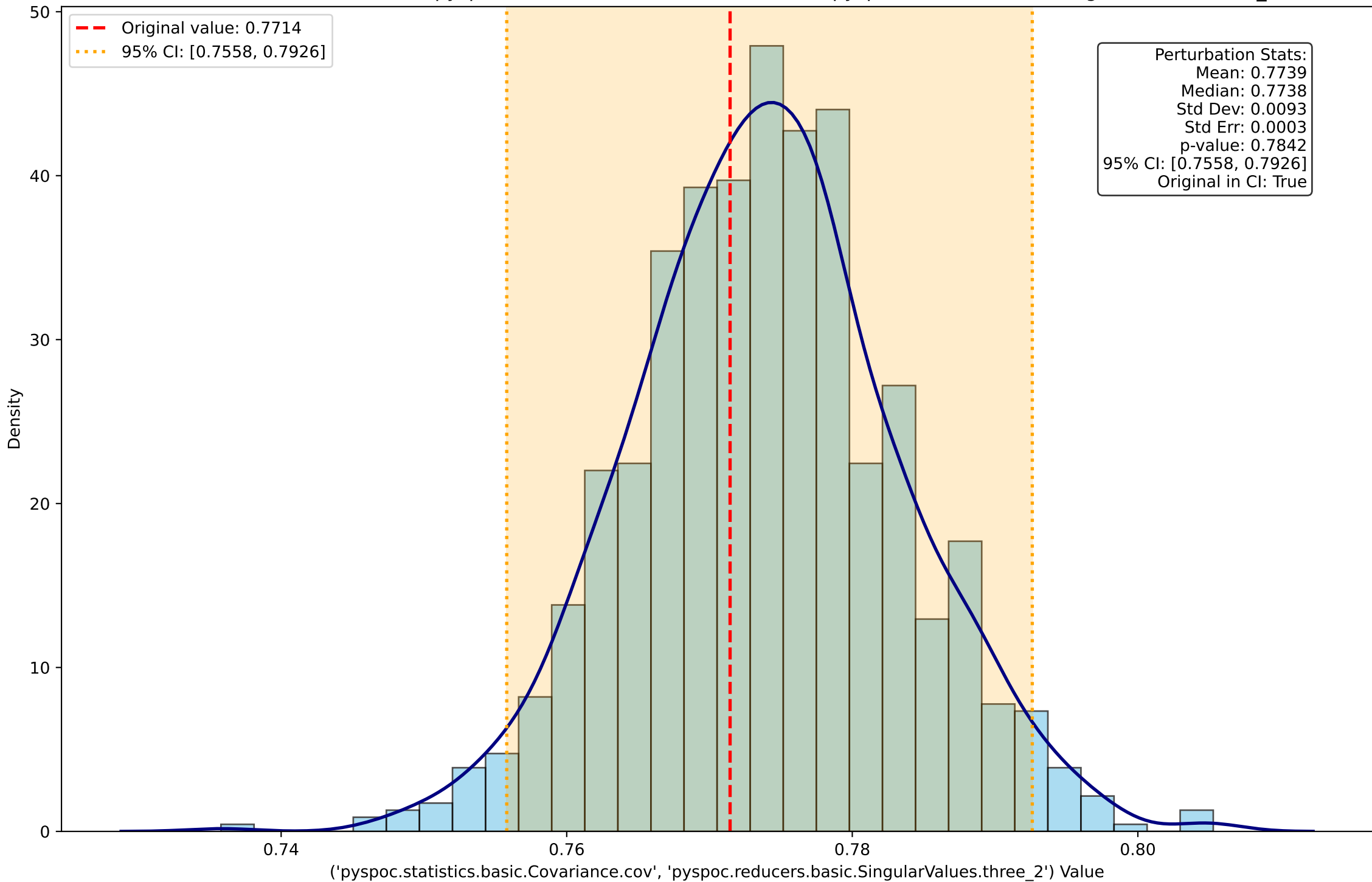
Column Bootstrap Distribution for ('pyspoc.statistics.basic.KendallTau.standard', 'pyspoc.reducers.basic.SingularValues.three_3')



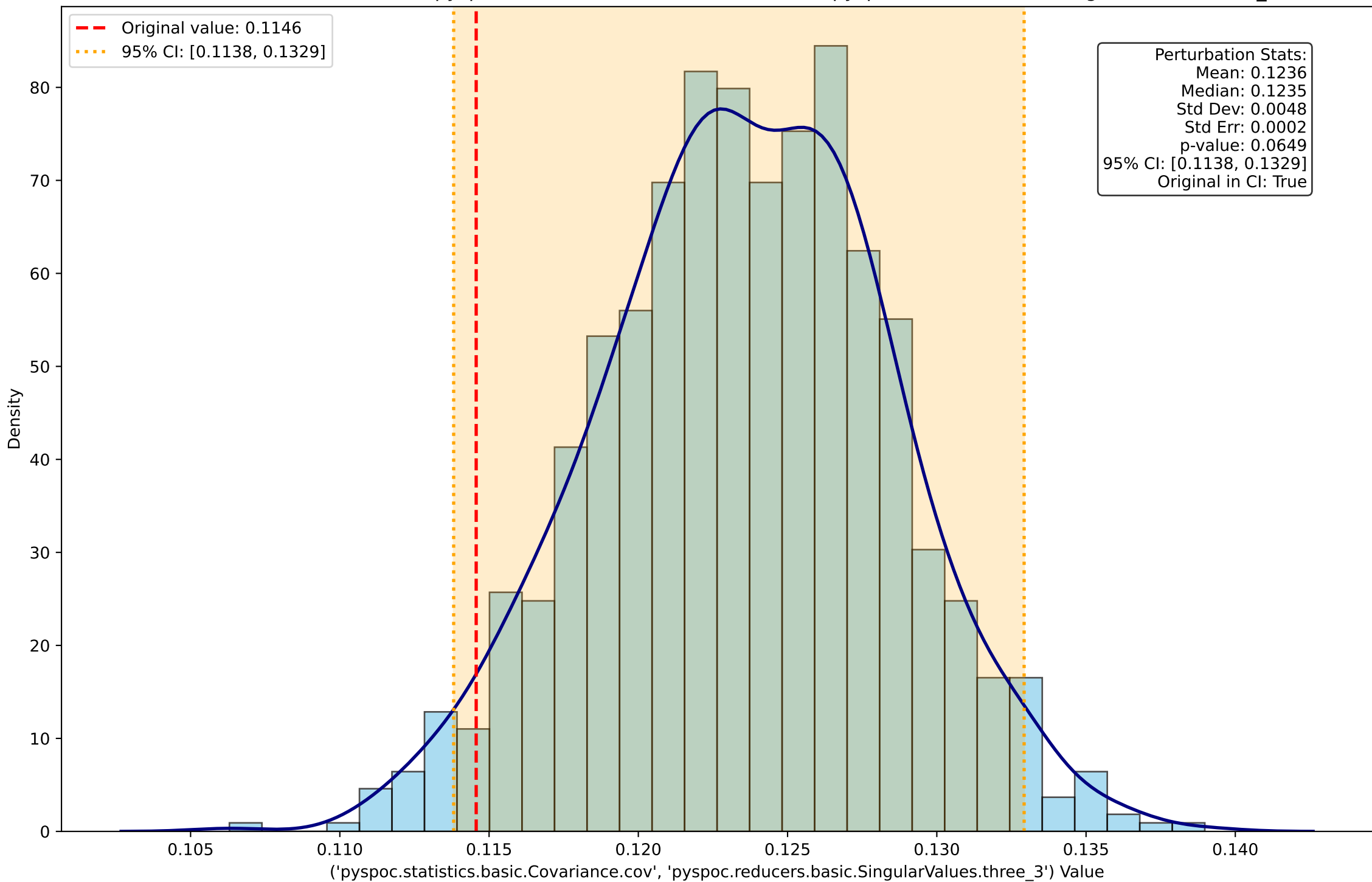
Perturbation Distribution for ('pypoc.statistics.basic.Covariance.cov', 'pypoc.reducers.basic.SingularValues.three_1')

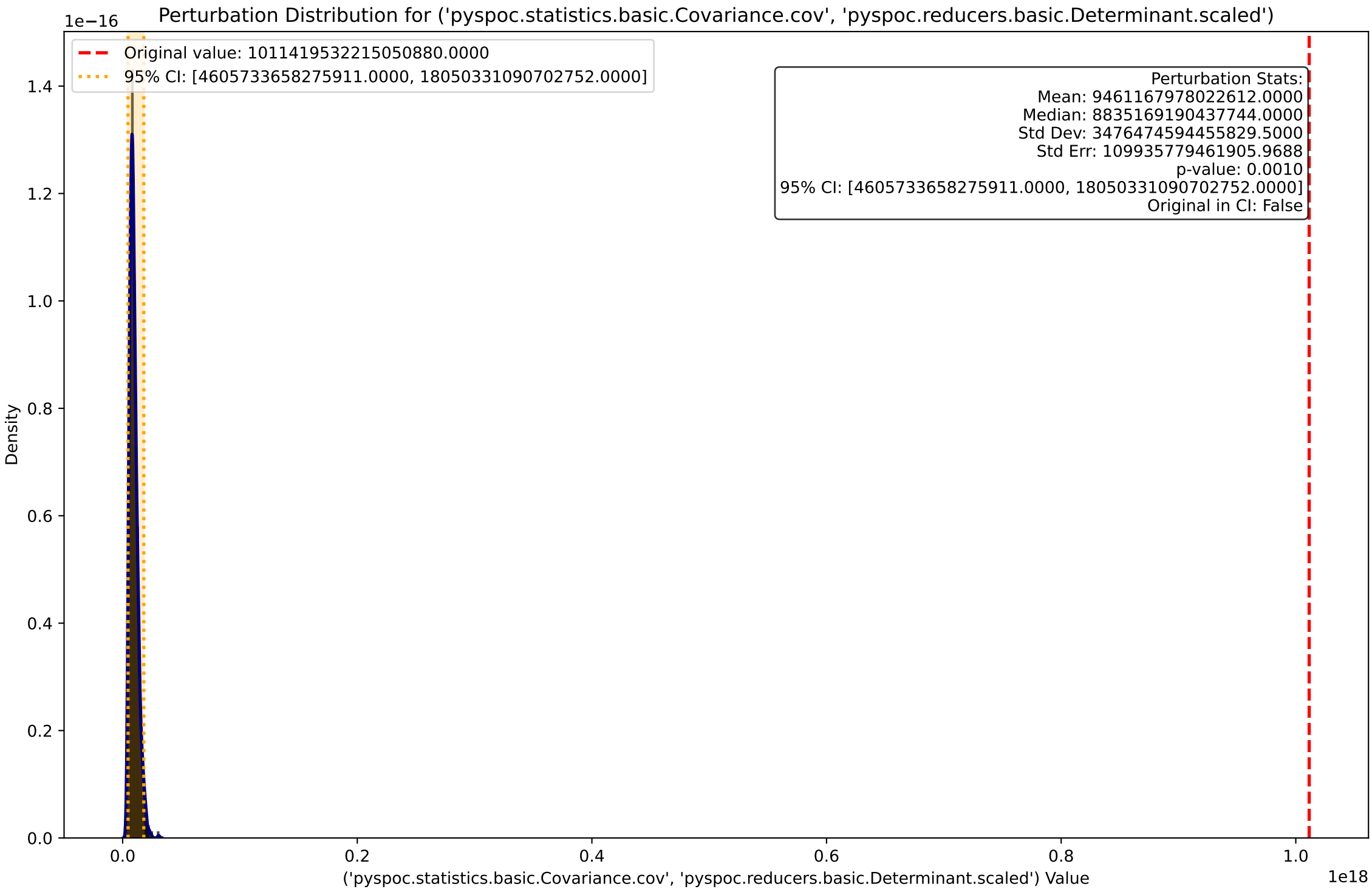


Perturbation Distribution for ('pypoc.statistics.basic.Covariance.cov', 'pypoc.reducers.basic.SingularValues.three_2')



Perturbation Distribution for ('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.SingularValues.three_3')





Perturbation Distribution for ('pypoc.statistics.basic.Covariance.cov', 'pypoc.reducers.basic.Determinant.non-scaled')

1e8

Original value: 0.0000
95% CI: [0.0000, 0.0000]

Perturbation Stats:
Mean: 0.0000
Median: 0.0000
Std Dev: 0.0000
Std Err: 0.0000
p-value: 0.0010
95% CI: [0.0000, 0.0000]
Original in CI: False

2.0

1.5

1.0

0.5

0.0

0.25

0.50

0.75

1.00

1.25

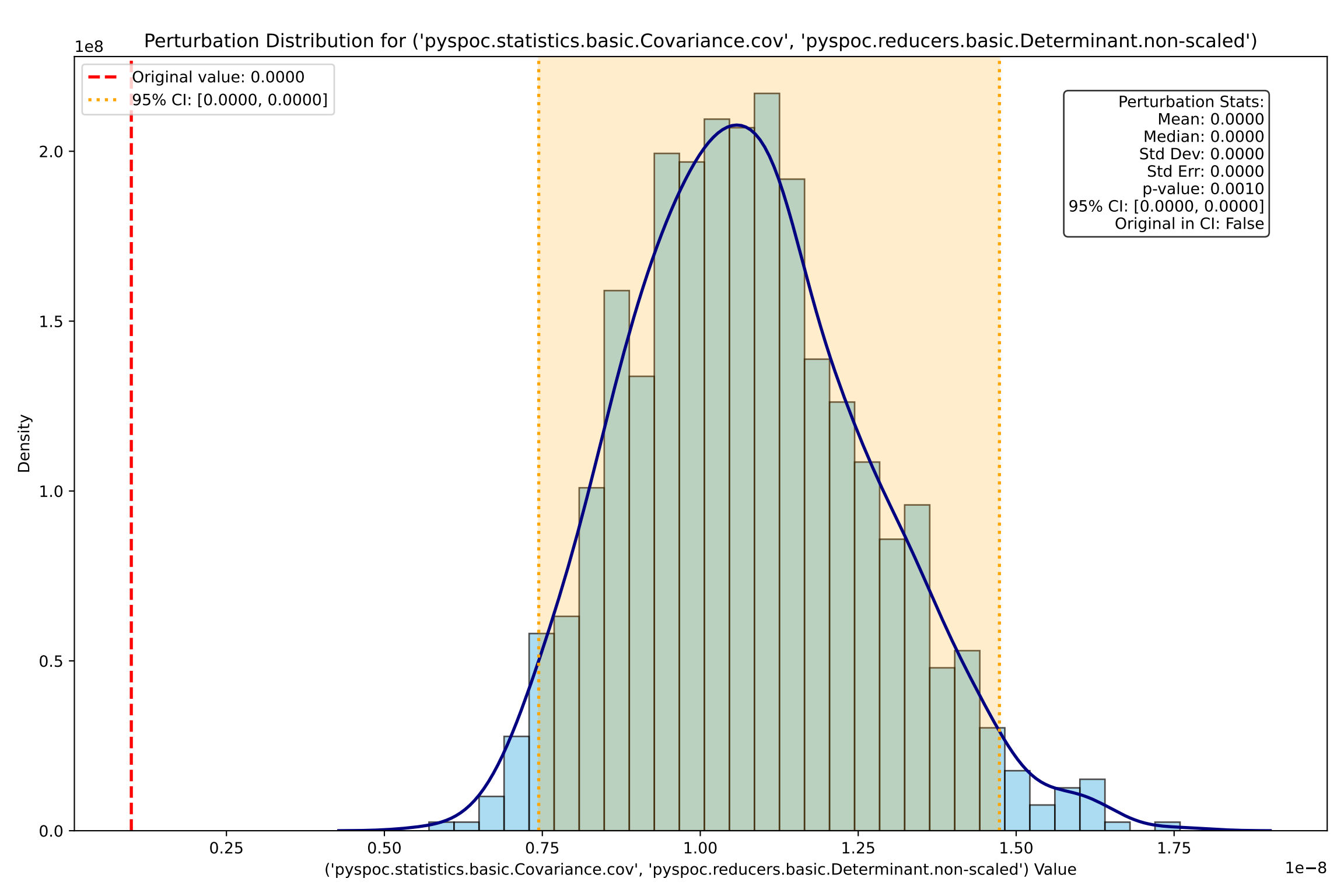
1.50

1.75

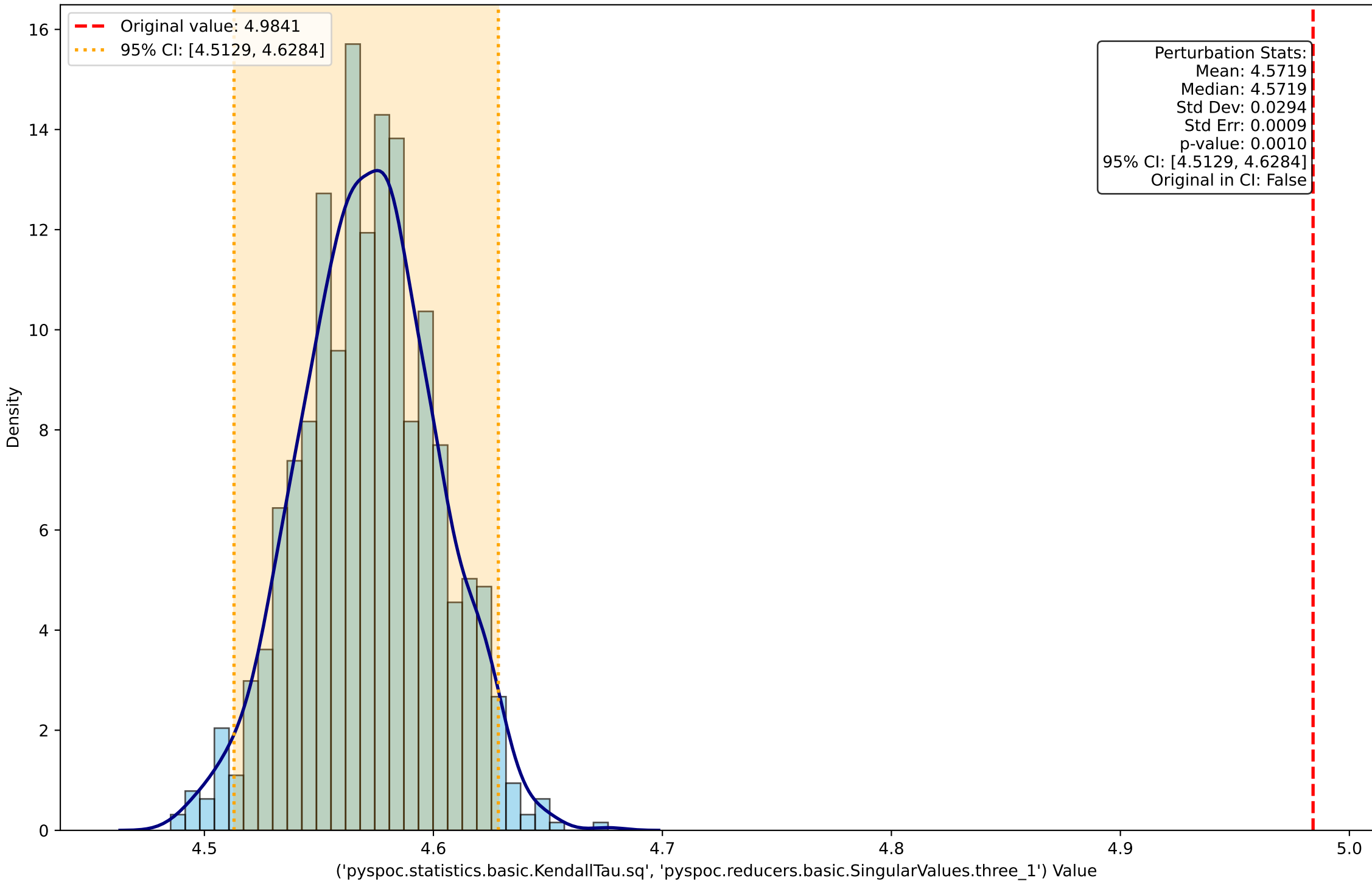
('pypoc.statistics.basic.Covariance.cov', 'pypoc.reducers.basic.Determinant.non-scaled') Value

1e-8

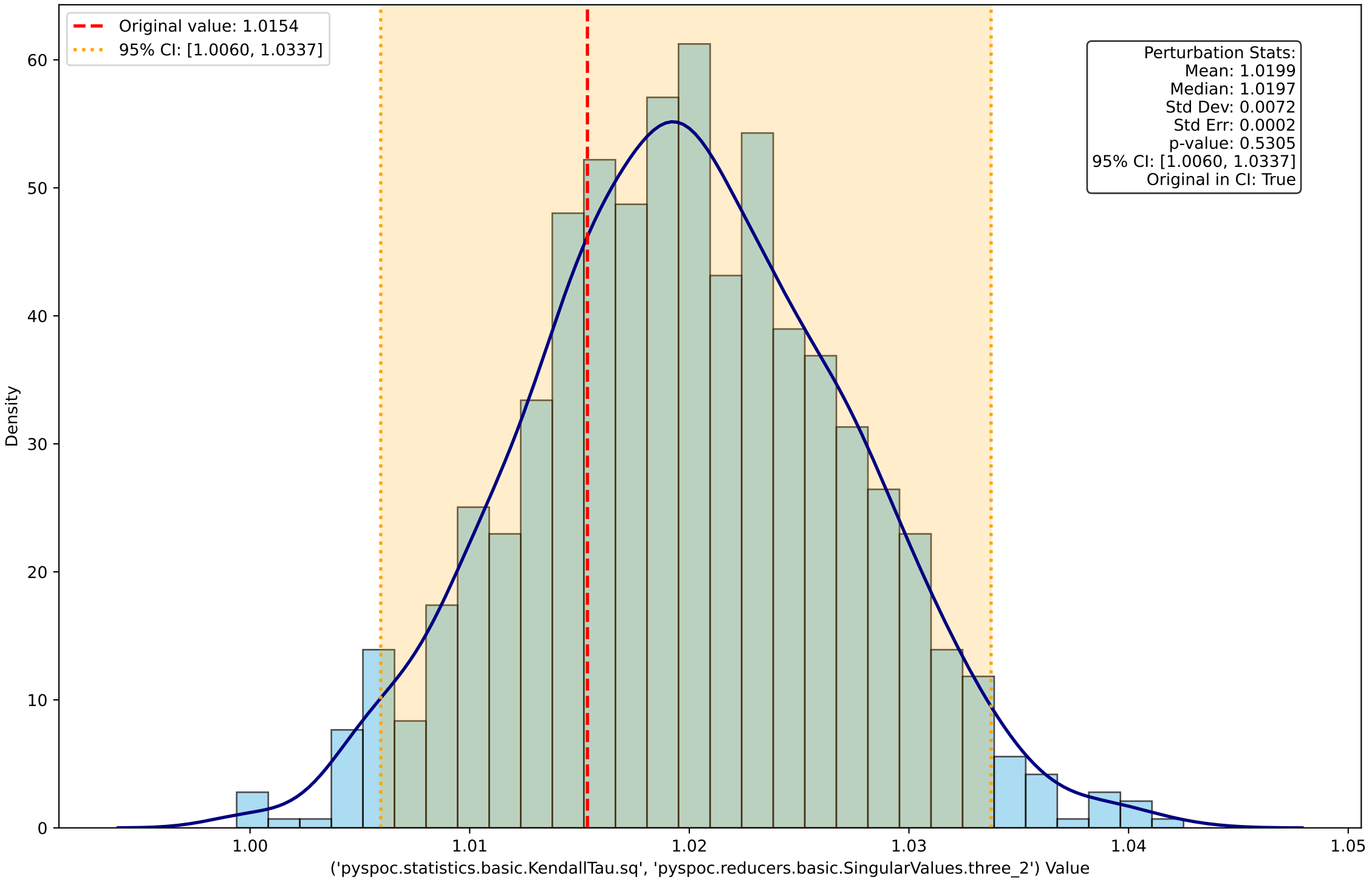
Density



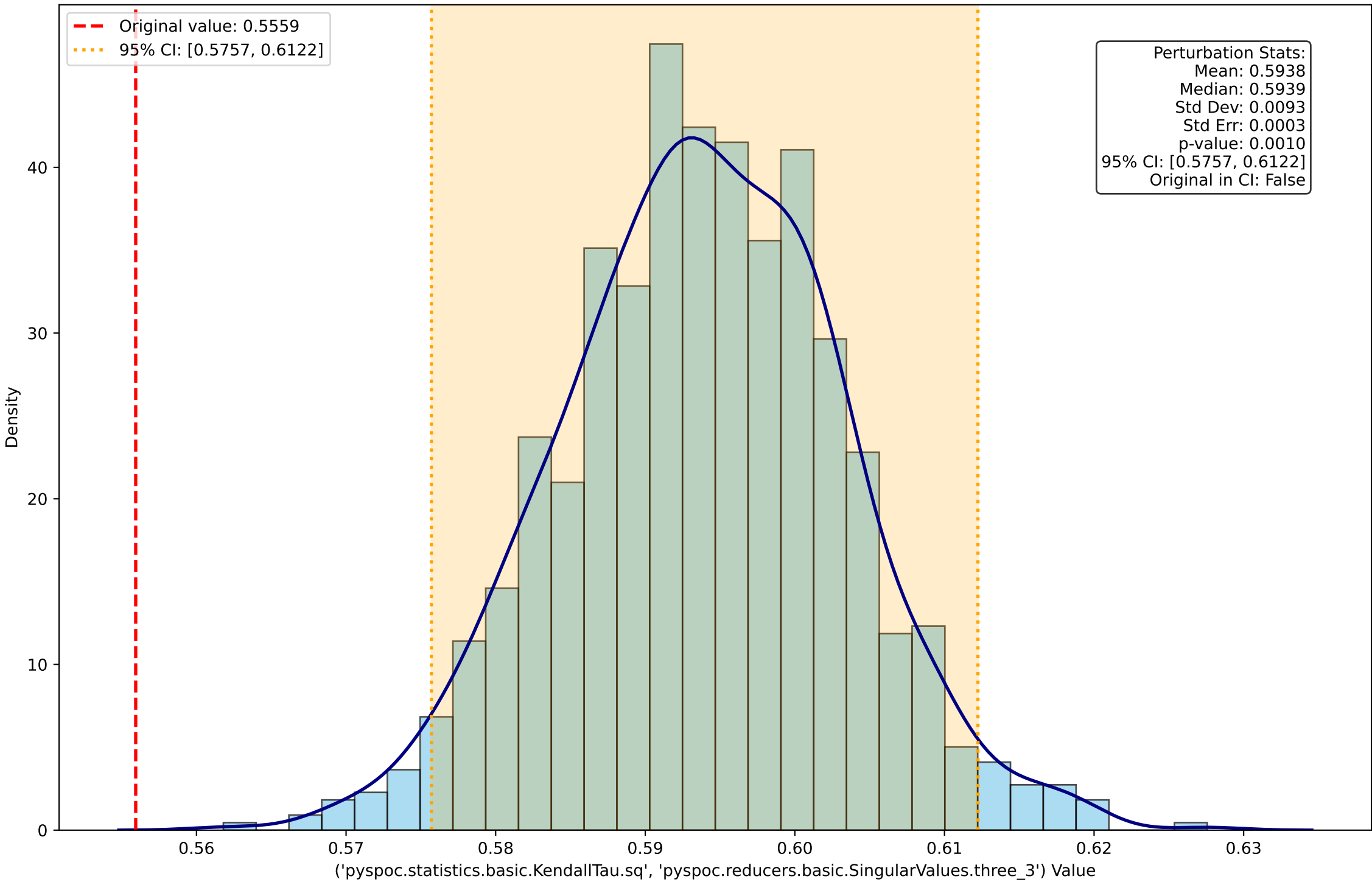
Perturbation Distribution for ('pypoc.statistics.basic.KendallTau.sq', 'pypoc.reducers.basic.SingularValues.three_1')



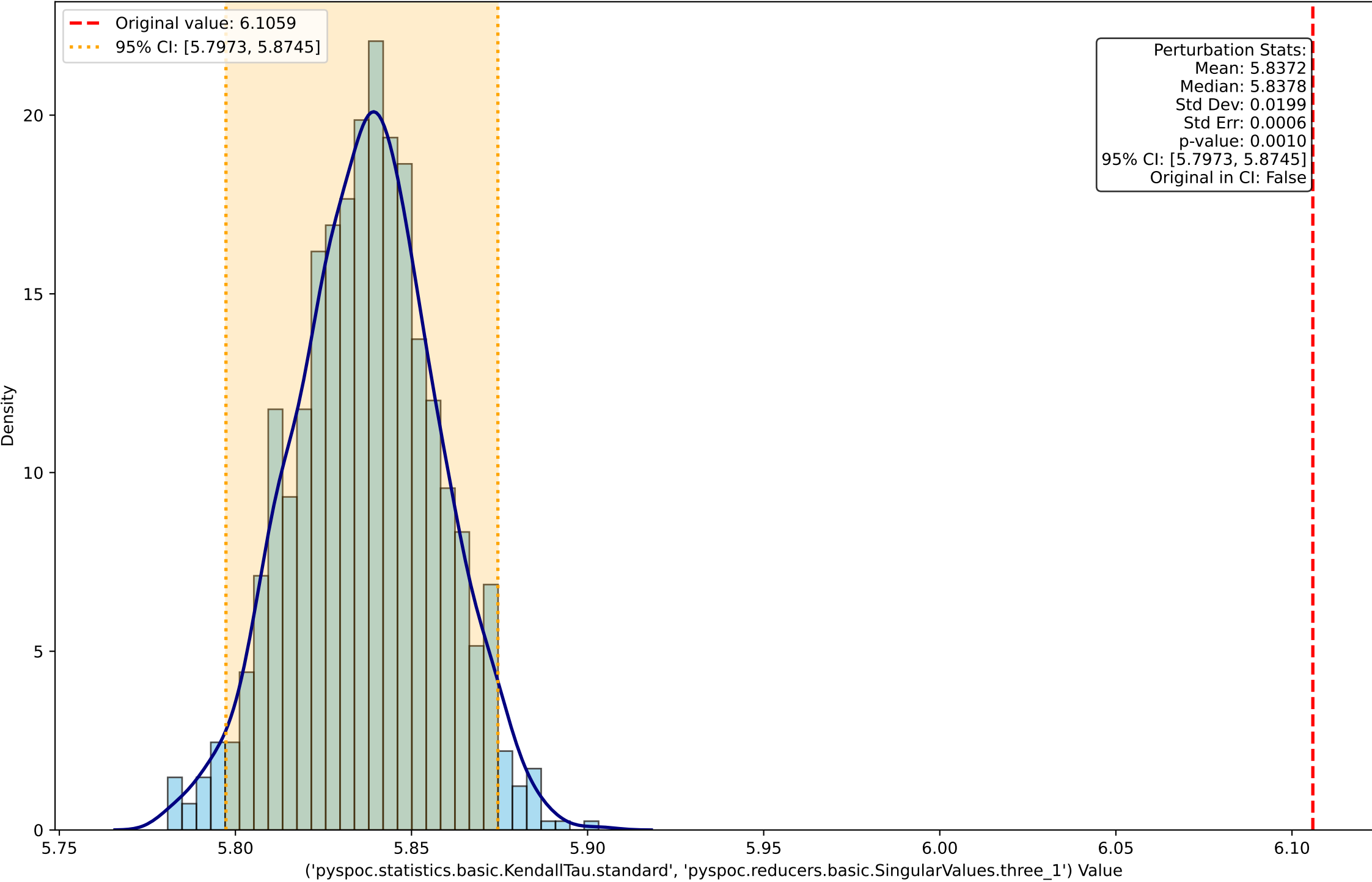
Perturbation Distribution for ('pyspoc.statistics.basic.KendallTau.sq', 'pyspoc.reducers.basic.SingularValues.three_2')



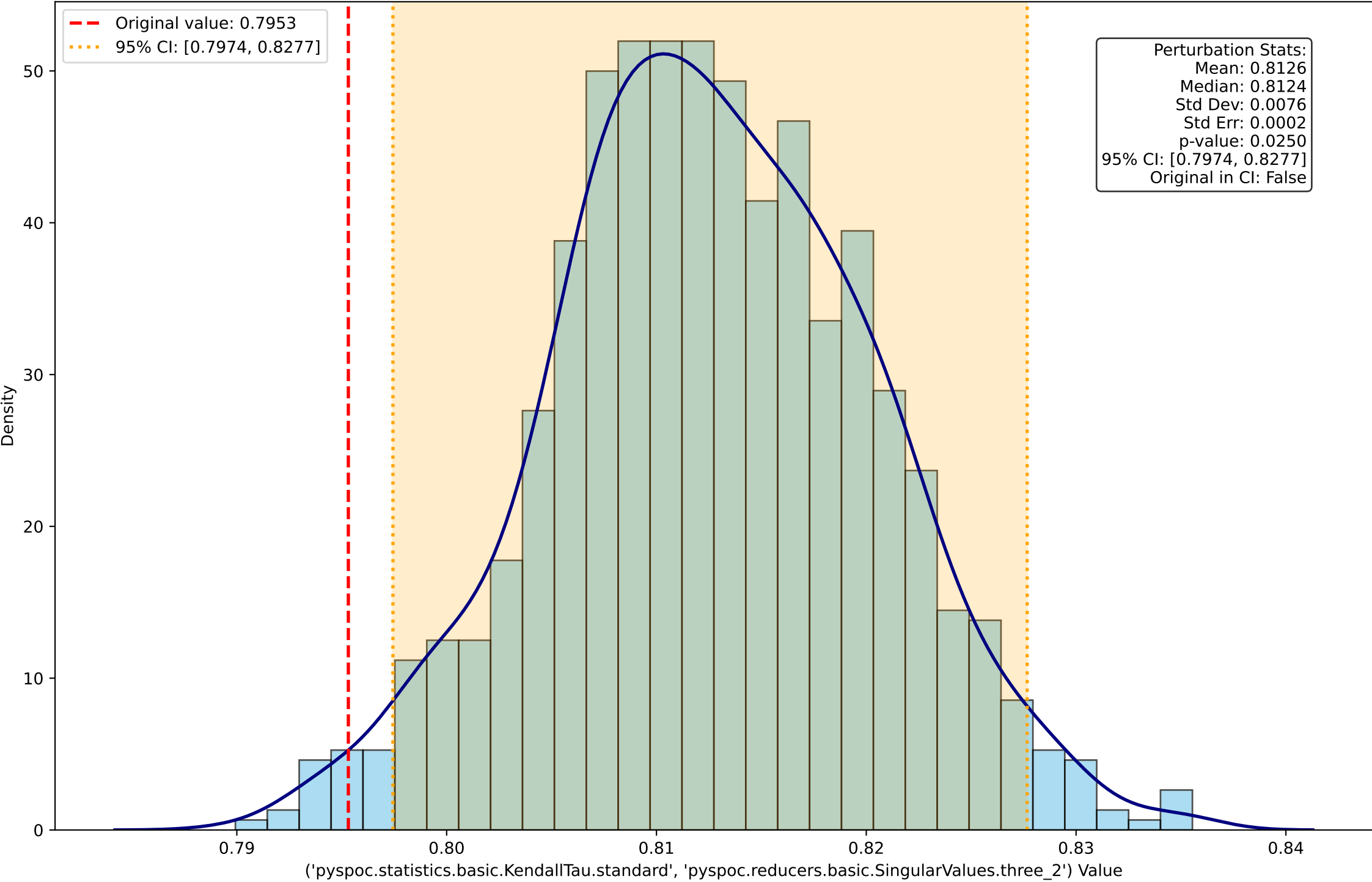
Perturbation Distribution for ('pyspoc.statistics.basic.KendallTau.sq', 'pyspoc.reducers.basic.SingularValues.three_3')



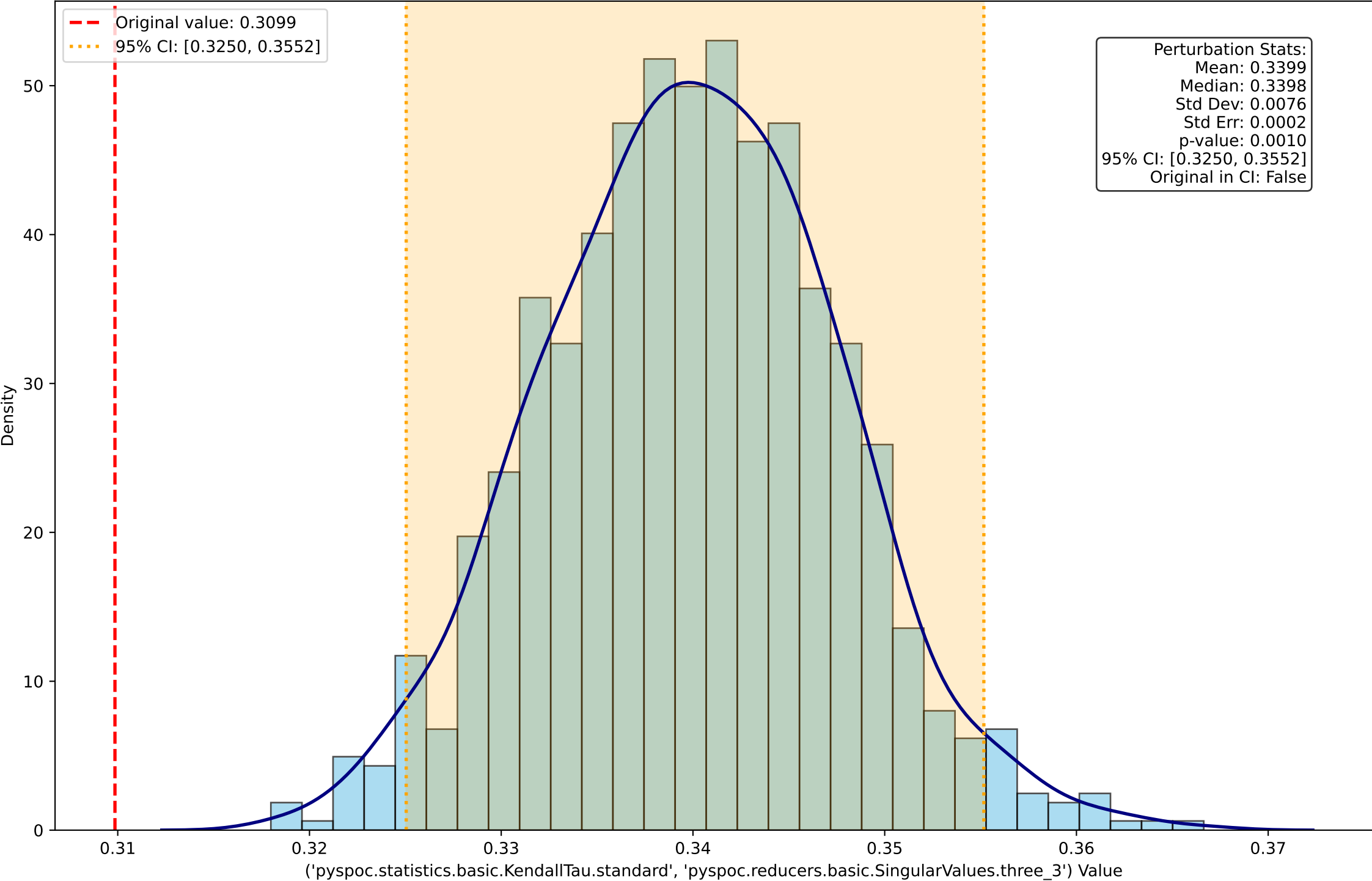
Perturbation Distribution for ('pyspoc.statistics.basic.KendallTau.standard', 'pyspoc.reducers.basic.SingularValues.three_1')



Perturbation Distribution for ('pyspoc.statistics.basic.KendallTau.standard', 'pyspoc.reducers.basic.SingularValues.three_2')



Perturbation Distribution for ('pyspoc.statistics.basic.KendallTau.standard', 'pyspoc.reducers.basic.SingularValues.three_3')



P-value Table for Each Reducer

Reducer	Permutation p-value	Row Bootstrap p-value	Col Bootstrap p-value	Perturbation p-value
('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.SingularValues.three_1')	0.001998	0.98501	0.001998	0.000999
('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.SingularValues.three_2')	0.001998	0.999	0.001998	0.78422
('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.SingularValues.three_3')	0.001998	0.93107	0.001998	0.064935
('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.Determinant.scaled')	0.001998	0.70529	0.001998	0.000999
('pyspoc.statistics.basic.Covariance.cov', 'pyspoc.reducers.basic.Determinant.non-scaled')	0.001998	0.70529	0.001998	0.000999
('pyspoc.statistics.basic.KendallTau.sq', 'pyspoc.reducers.basic.SingularValues.three_1')	0.001998	0.93506	0.001998	0.000999
('pyspoc.statistics.basic.KendallTau.sq', 'pyspoc.reducers.basic.SingularValues.three_2')	0.05994	0.96903	0.001998	0.53047
('pyspoc.statistics.basic.KendallTau.sq', 'pyspoc.reducers.basic.SingularValues.three_3')	0.001998	0.84715	0.001998	0.000999
('pyspoc.statistics.basic.KendallTau.standard', 'pyspoc.reducers.basic.SingularValues.three_1')	0.001998	0.999	0.001998	0.000999
('pyspoc.statistics.basic.KendallTau.standard', 'pyspoc.reducers.basic.SingularValues.three_2')	0.001998	0.95704	0.001998	0.024975
('pyspoc.statistics.basic.KendallTau.standard', 'pyspoc.reducers.basic.SingularValues.three_3')	0.001998	0.73127	0.001998	0.000999